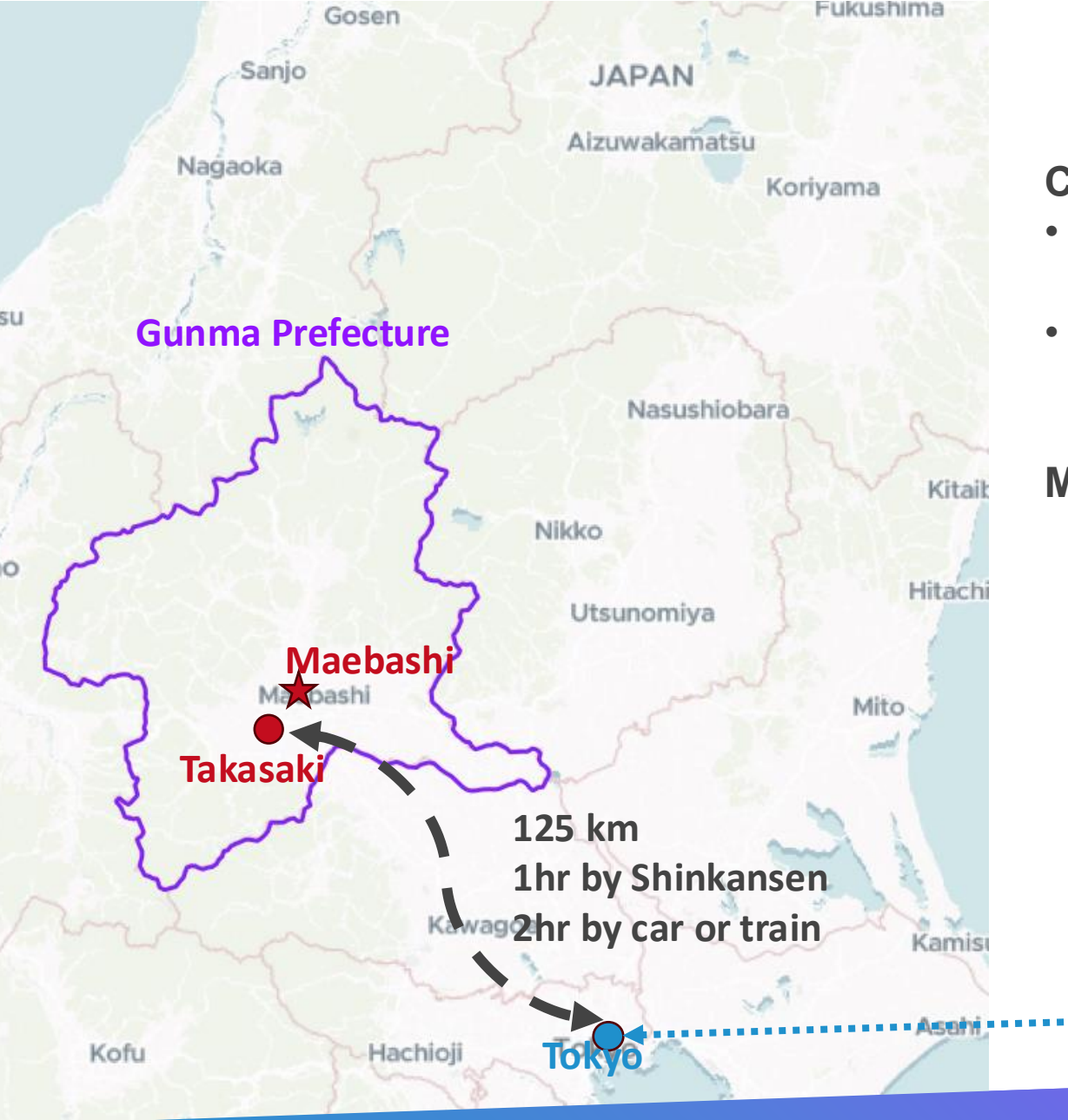


# Introducing DRT and Person-Based Accessibility in MATSim

*MATSim User Meeting 2026 | Paris, France*

**Jakob Rehmann**, Lichen Luo, Kai Nagel | September 28<sup>th</sup>, 2026



## Context:

- Transport disadvantage in rural areas, especially among the elderly
- Demand Responsive Transport (DRT)'s potential to improve equity

## Methods: Accessibility in MATSim

1. DRT Accessibility
2. Person-Based Accessibility



# Accessibility

= “the extent to which **land-use** and **transport** systems enable (groups of) **individuals** to reach activities or destinations by means of a (combination of) transport mode(s)” (Geurs & van Wee, 2004)

## Utility-Based Accessibility:

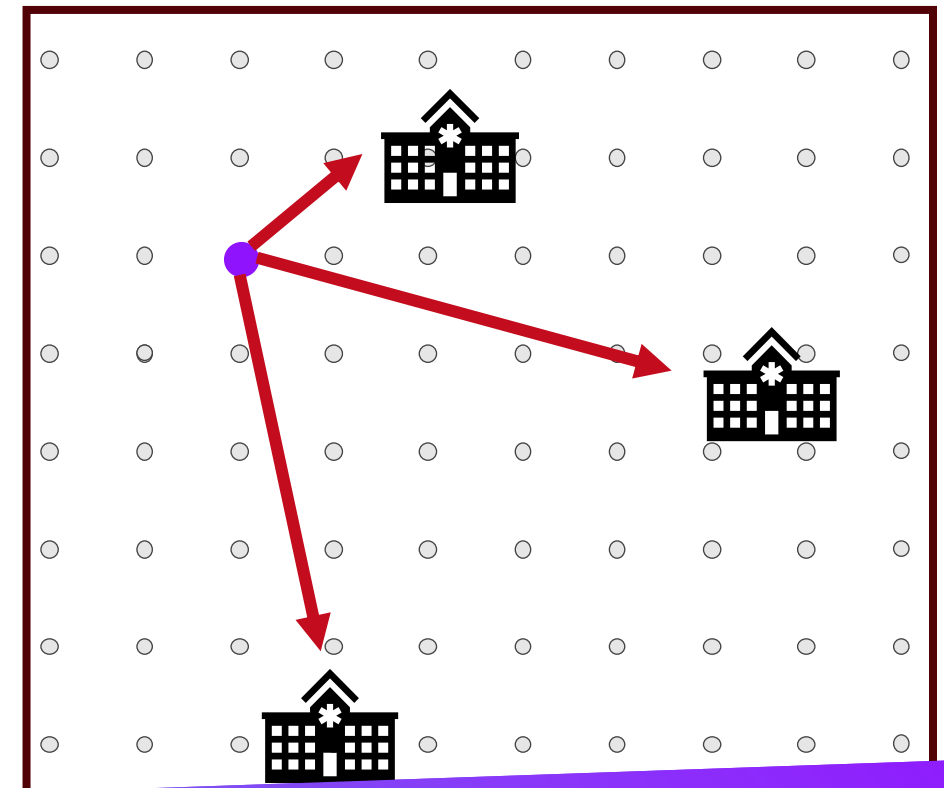
(Ben-Akiva & Lerman, 1985)

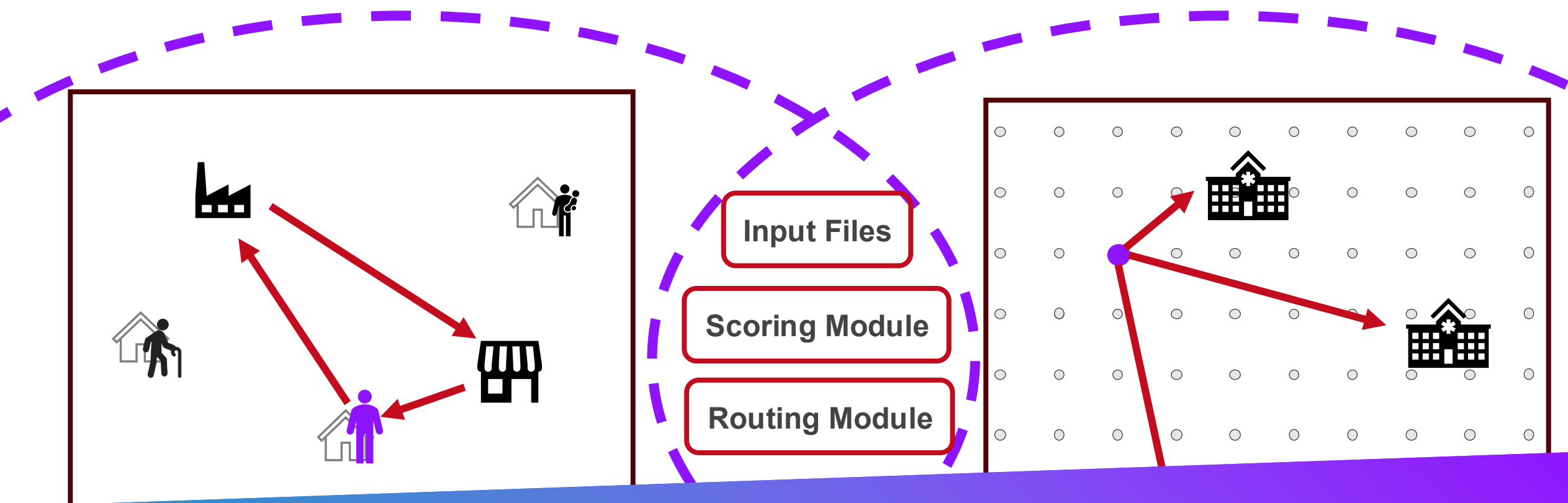
$$A_{\ell} := \ln \sum_k e^{V_{\ell k}}$$

Utility for travel from measuring point  $\ell$  to opportunity  $k$

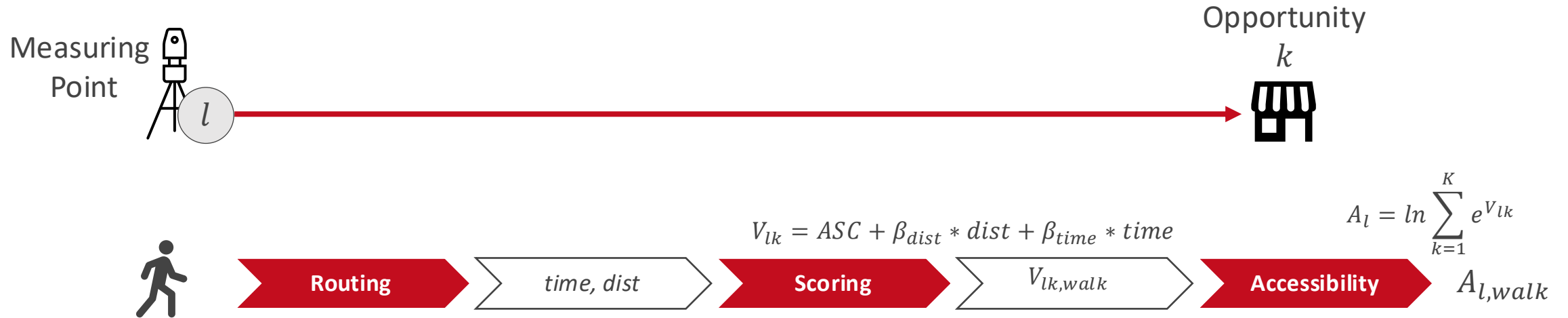
### Components of Accessibility

1. Land-Use
2. Transport
3. Individual
4. Temporal

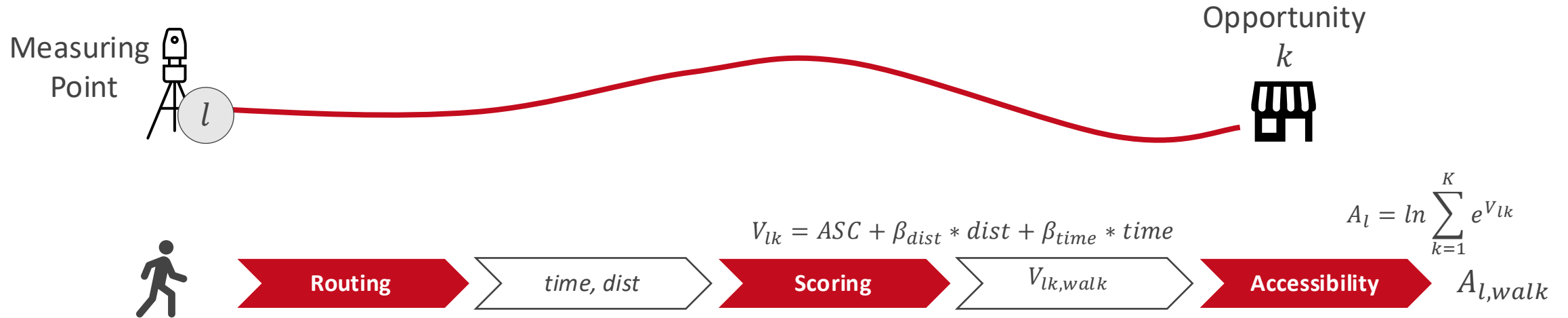




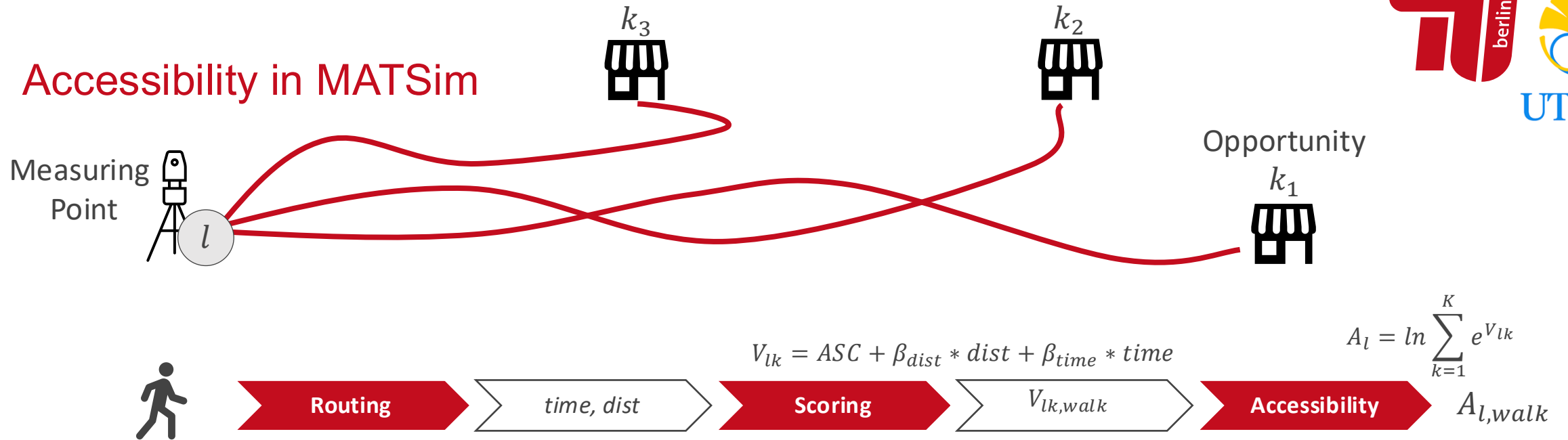
# Accessibility in MATSim



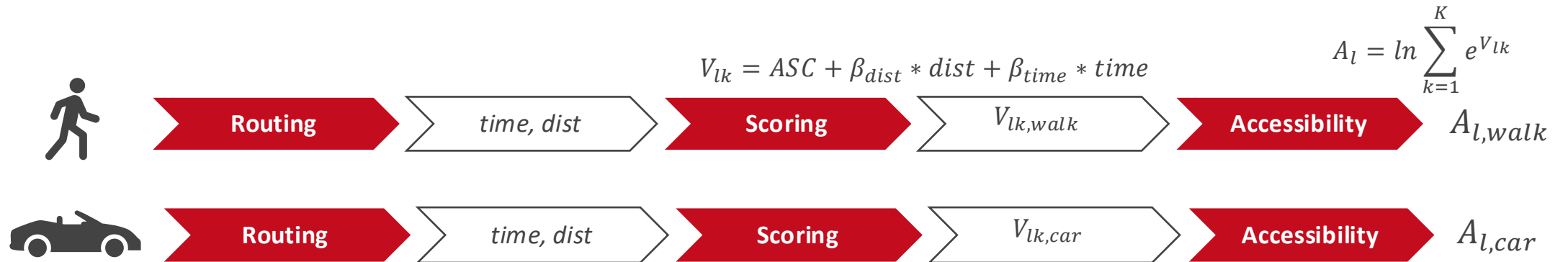
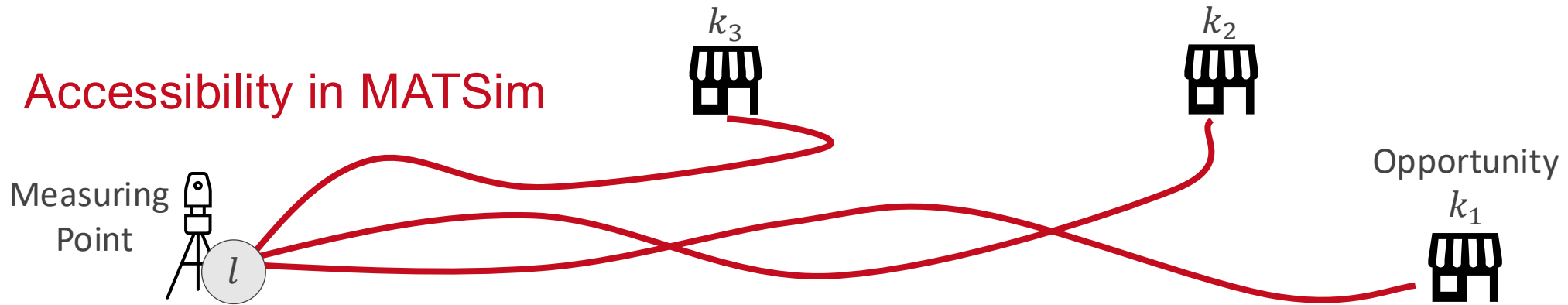
# Accessibility in MATSim



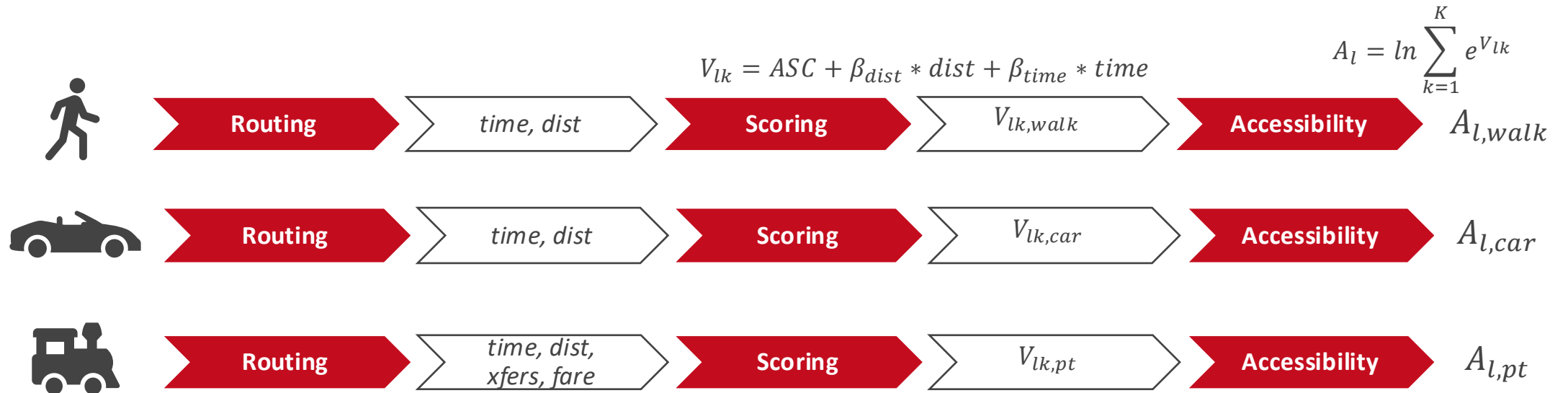
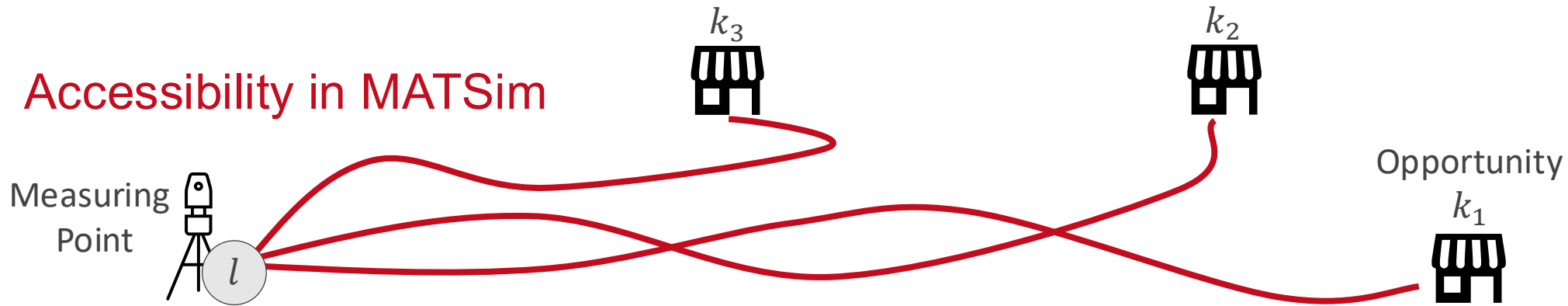
# Accessibility in MATSim



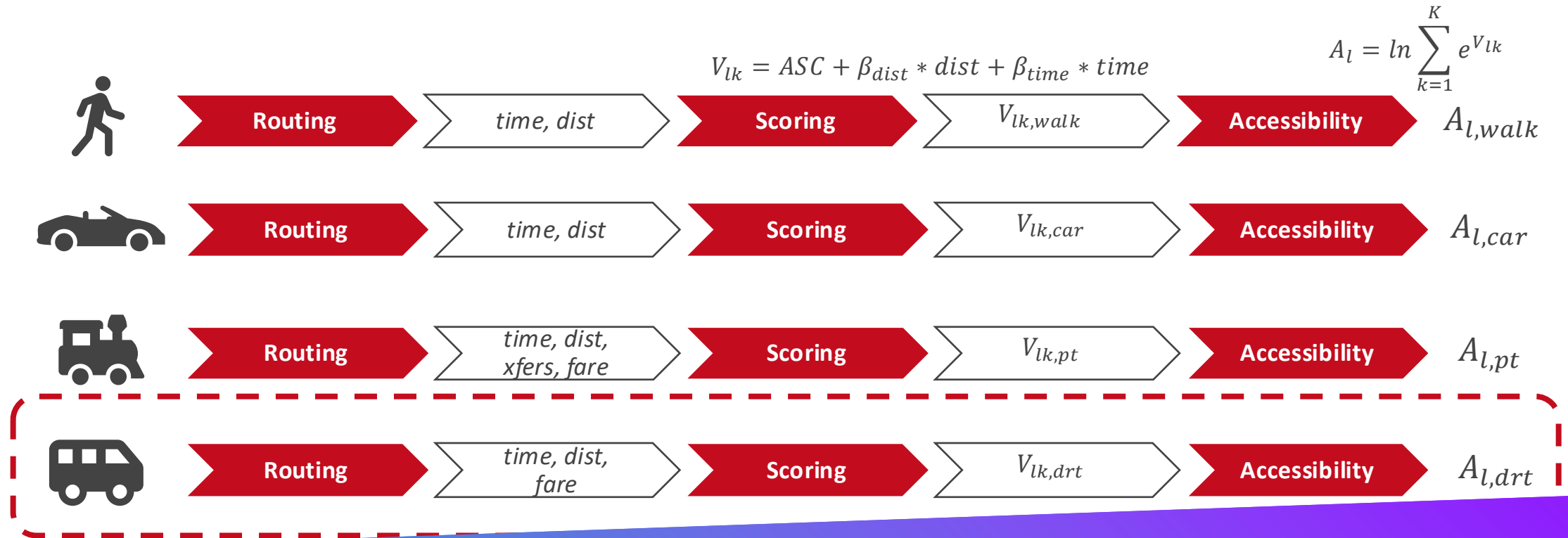
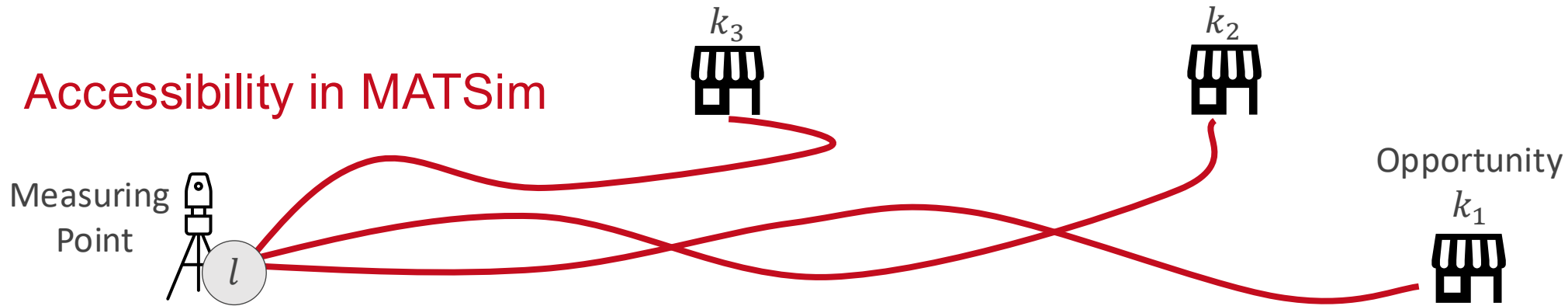
# Accessibility in MATSim



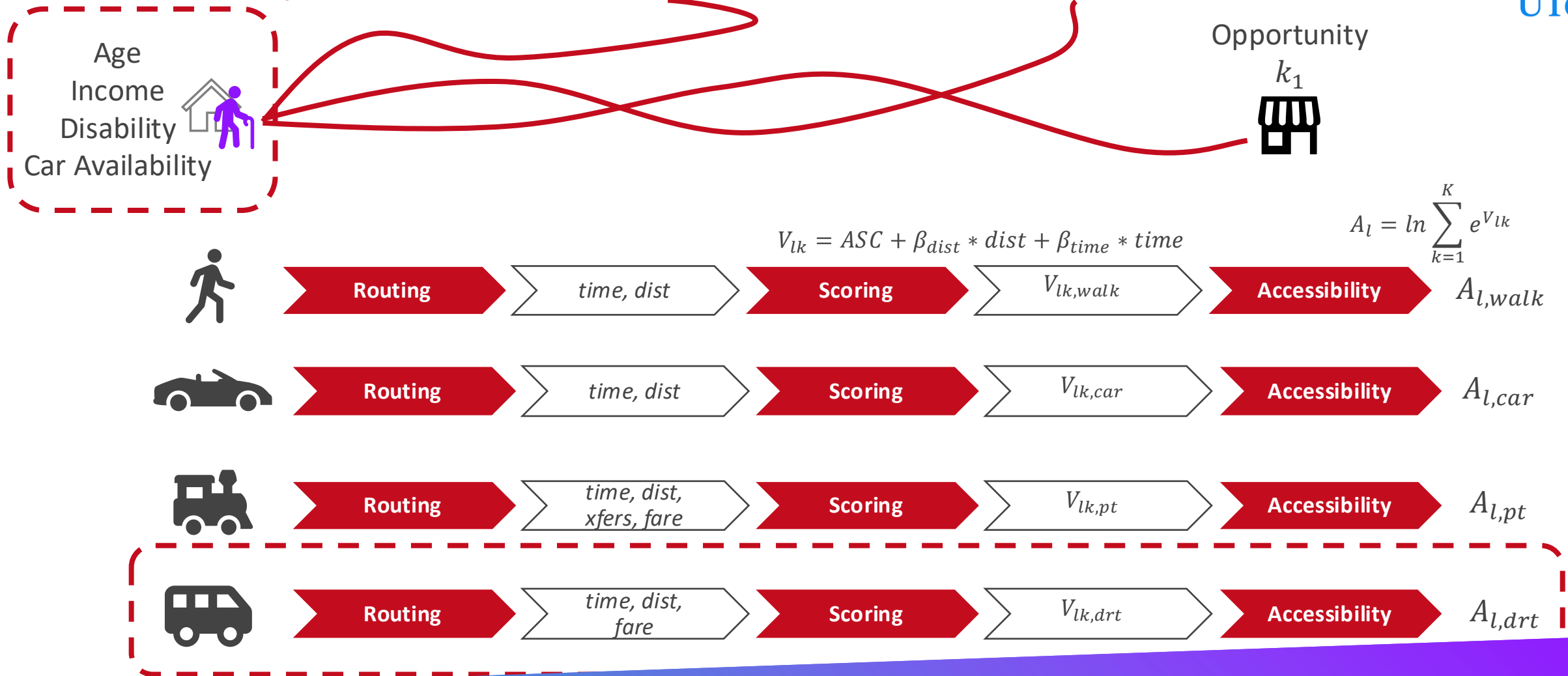
# Accessibility in MATSim



# Accessibility in MATSim



# Accessibility in MATSim



# DRT Accessibility

Introduction



DRT  
Accessibility

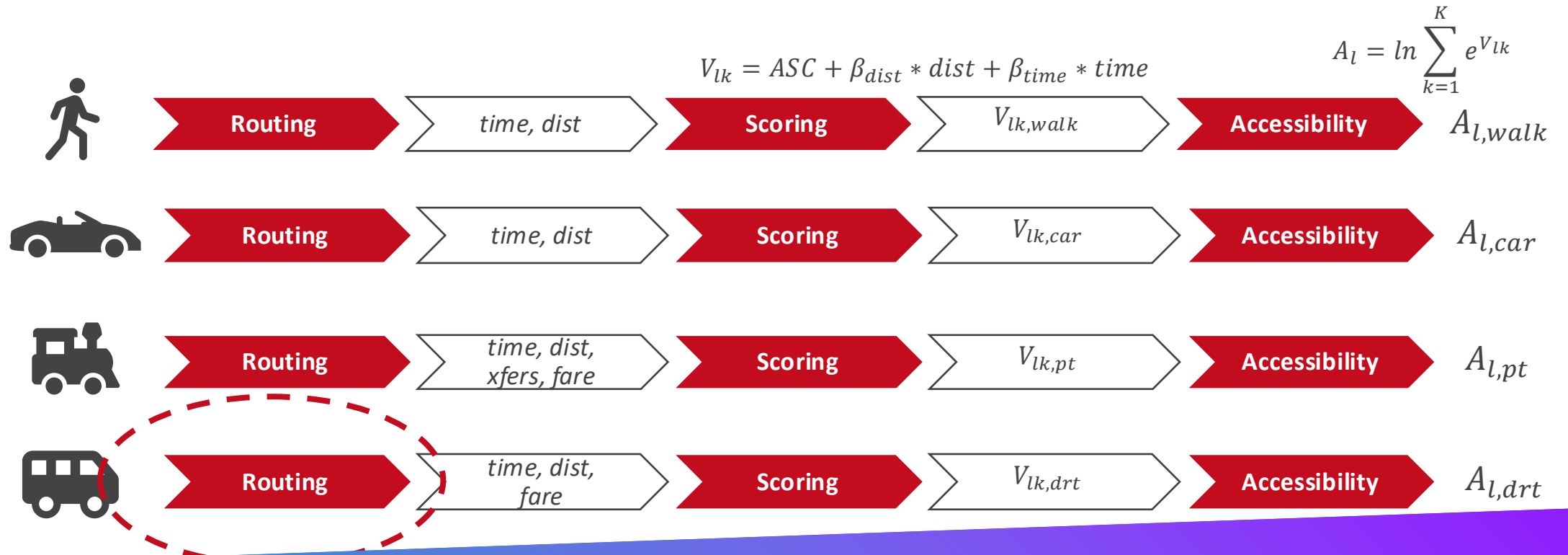
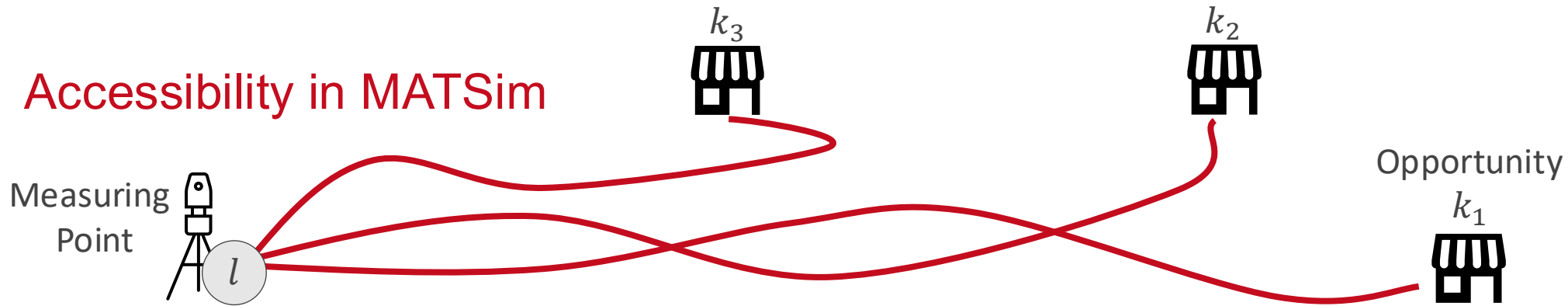


Person-based  
Accessibility

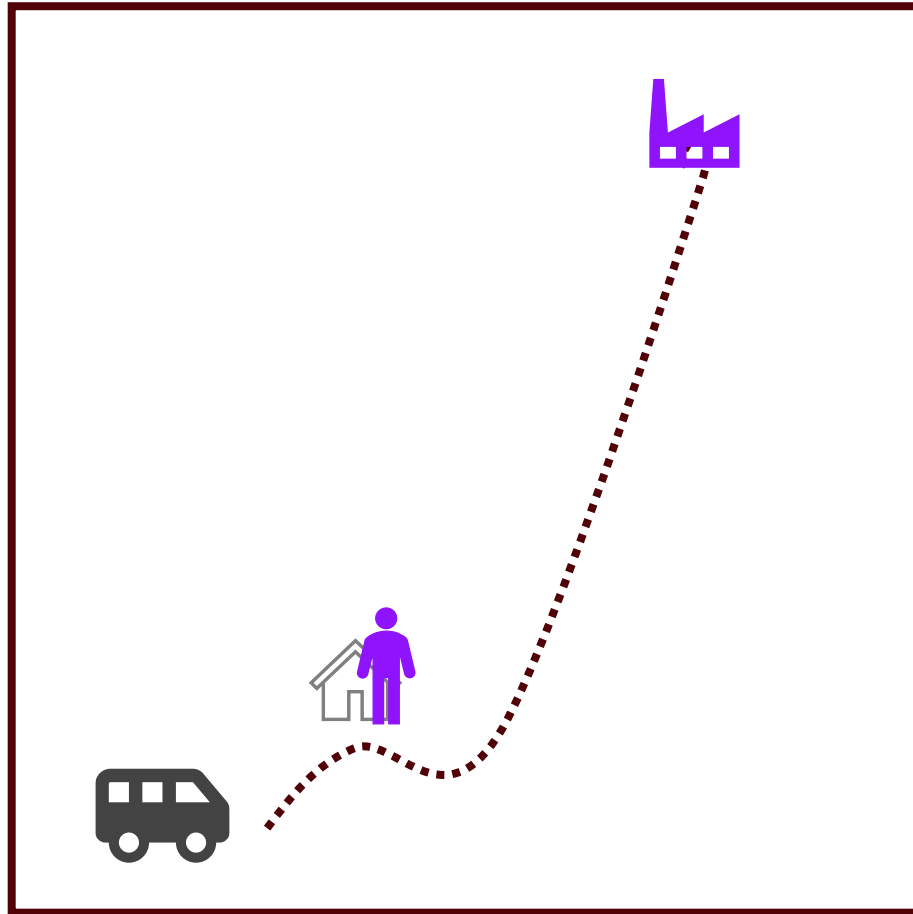


Discussion

# Accessibility in MATSim



## DRT Level of Service



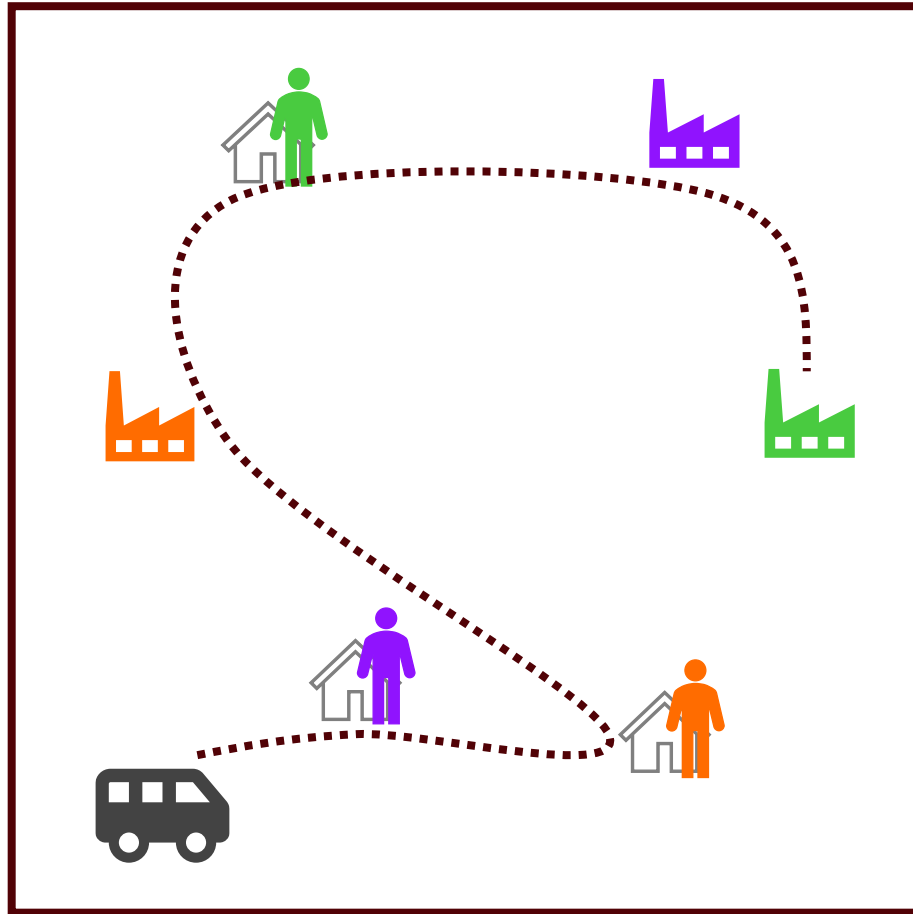
DRT Level of Service =

1. **Wait Time** until vehicle arrives
2. **Detour Factor** to pick up or drop off other passengers

These depend on:

1. **Demand:** number of ride requests, and spatial distribution of origins/destinations.
2. **Supply:** fleet size and spatial distribution of DRT vehicles

## DRT Level of Service



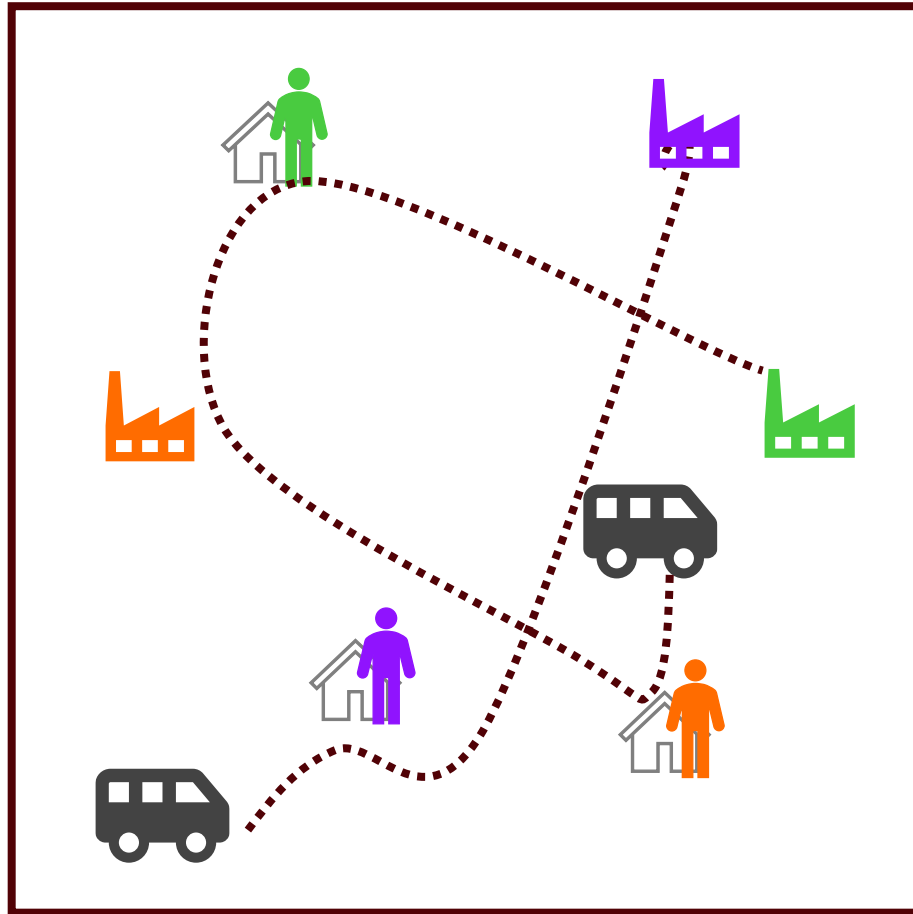
DRT Level of Service =

1. **Wait Time** until vehicle arrives
2. **Detour Factor** to pick up or drop off other passengers

These depend on:

1. **Demand:** number of ride requests, and spatial distribution of origins/destinations.
2. **Supply:** fleet size and spatial distribution of DRT vehicles

## DRT Level of Service



DRT Level of Service =

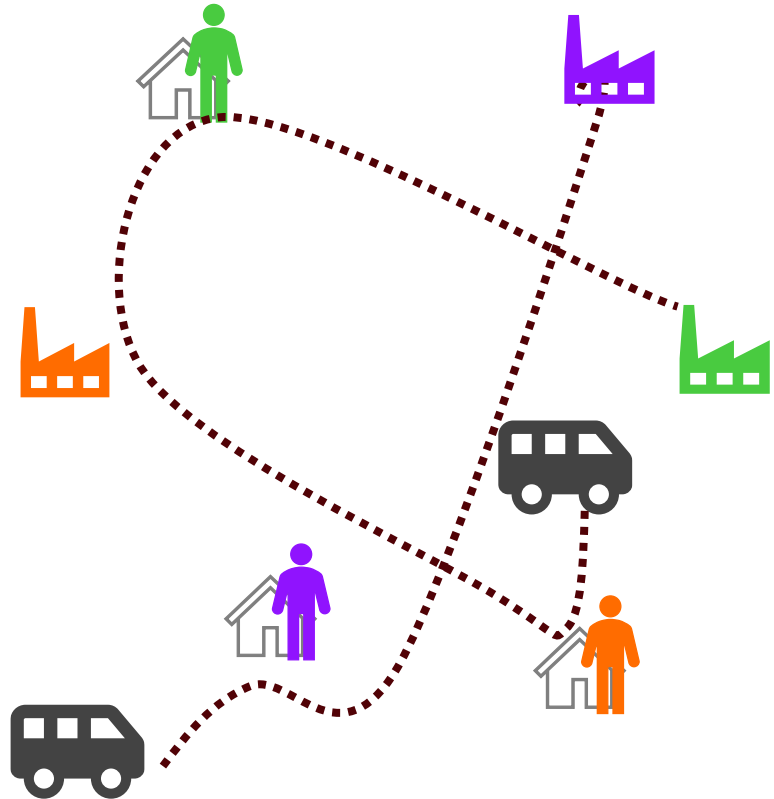
1. **Wait Time** until vehicle arrives
2. **Detour Factor** to pick up or drop off other passengers

These depend on:

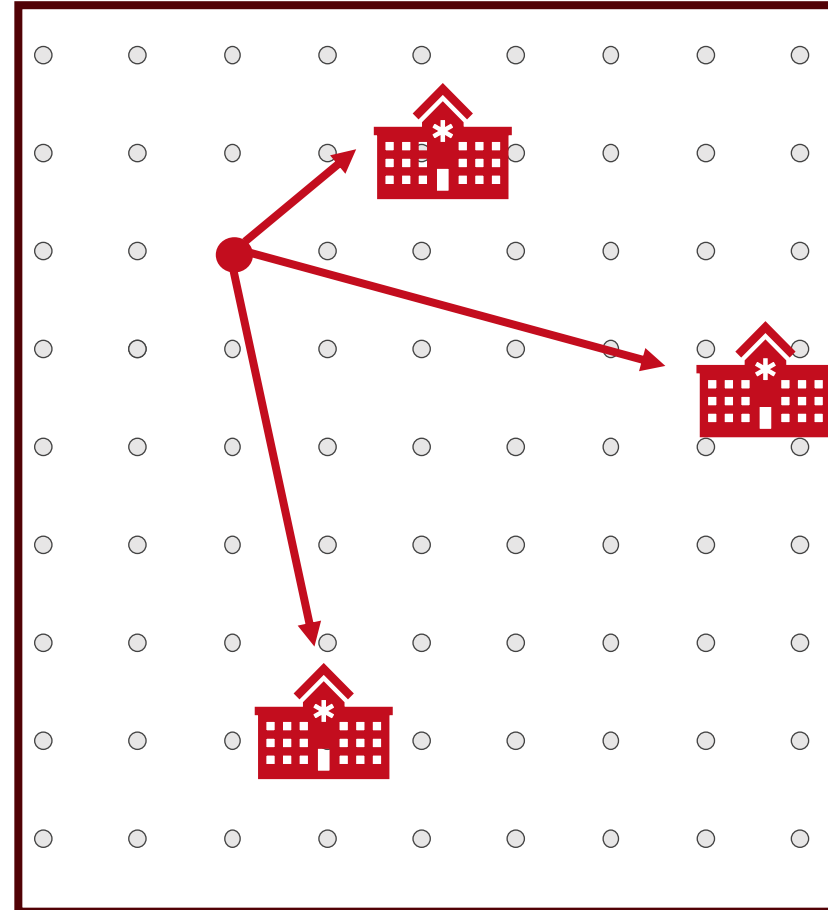
1. **Demand:** number of ride requests, and spatial distribution of origins/destinations.
2. **Supply:** fleet size and spatial distribution of DRT vehicles

# DRT Accessibility Challenge

“Realistic” Demand for DRT

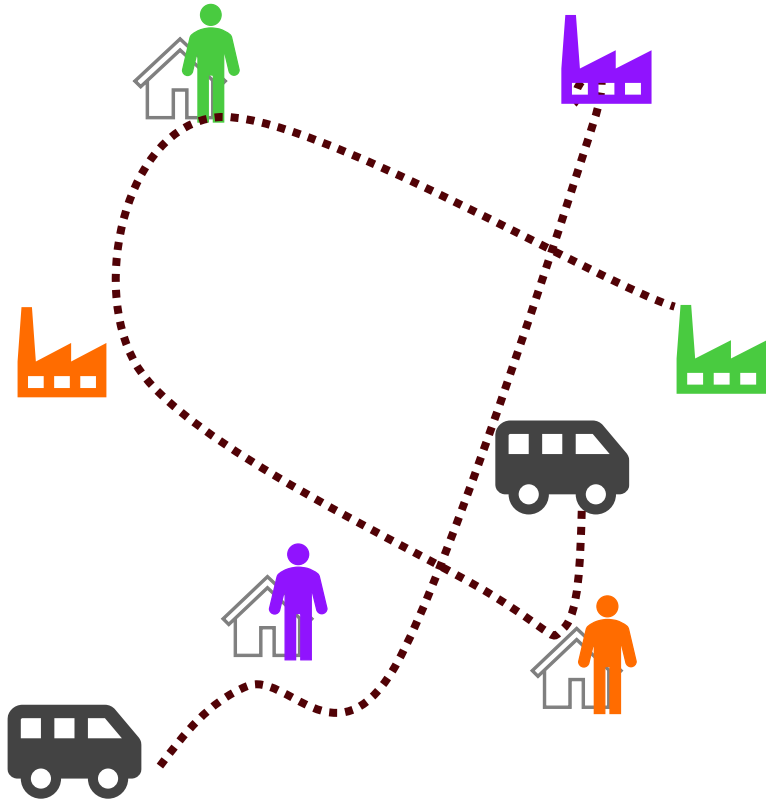


“Unrealistic” Demand needed for Accessibility Calculation

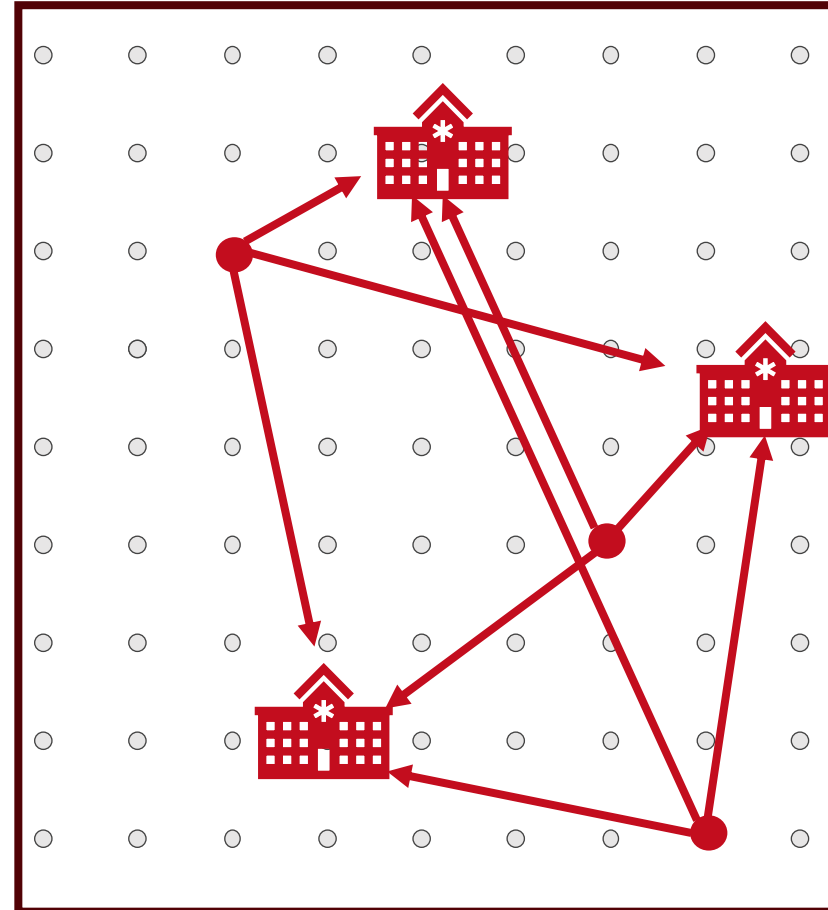


# DRT Accessibility Challenge

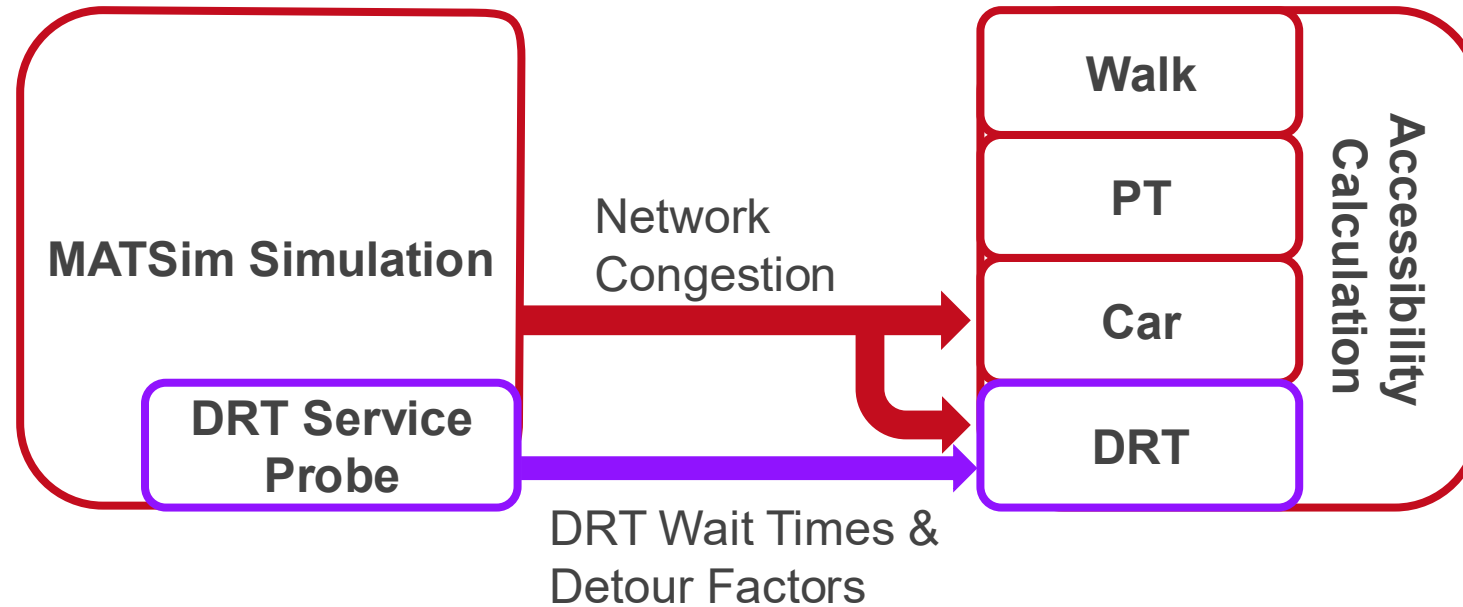
“Realistic” Demand for DRT



“Unrealistic” Demand needed for Accessibility Calculation

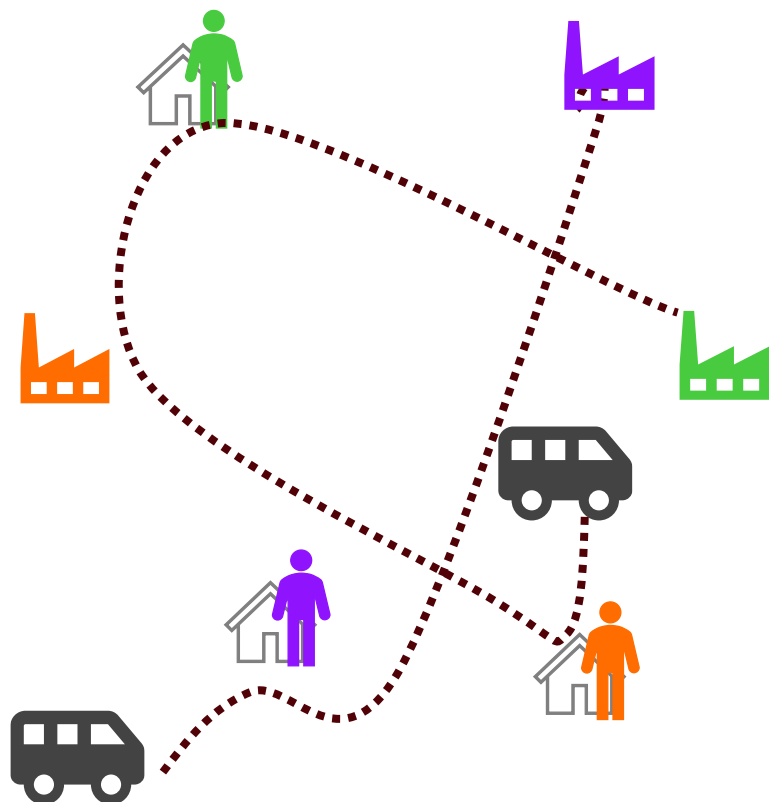


# DRT Accessibility Solution

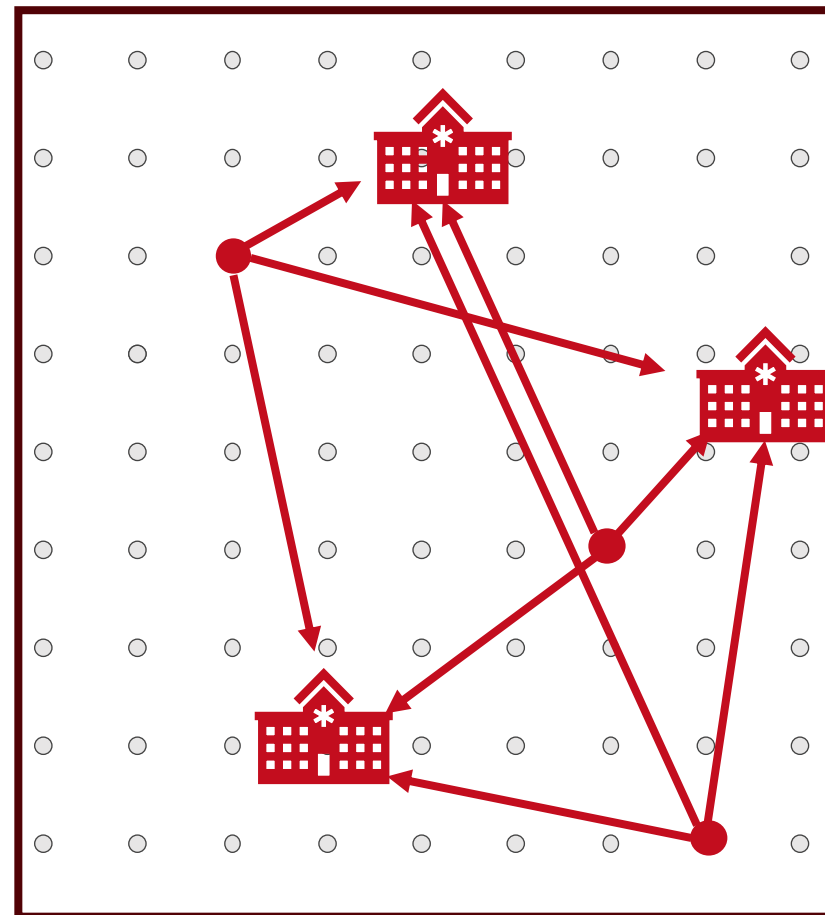


# Observational DRT Service Probe

“Realistic” Demand for DRT

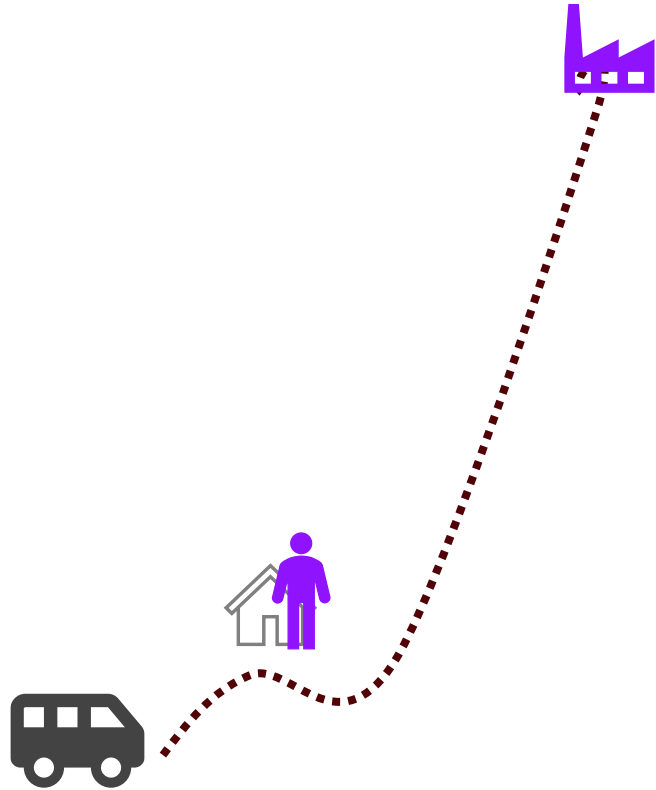


“Unrealistic” Demand needed for Accessibility Calculation

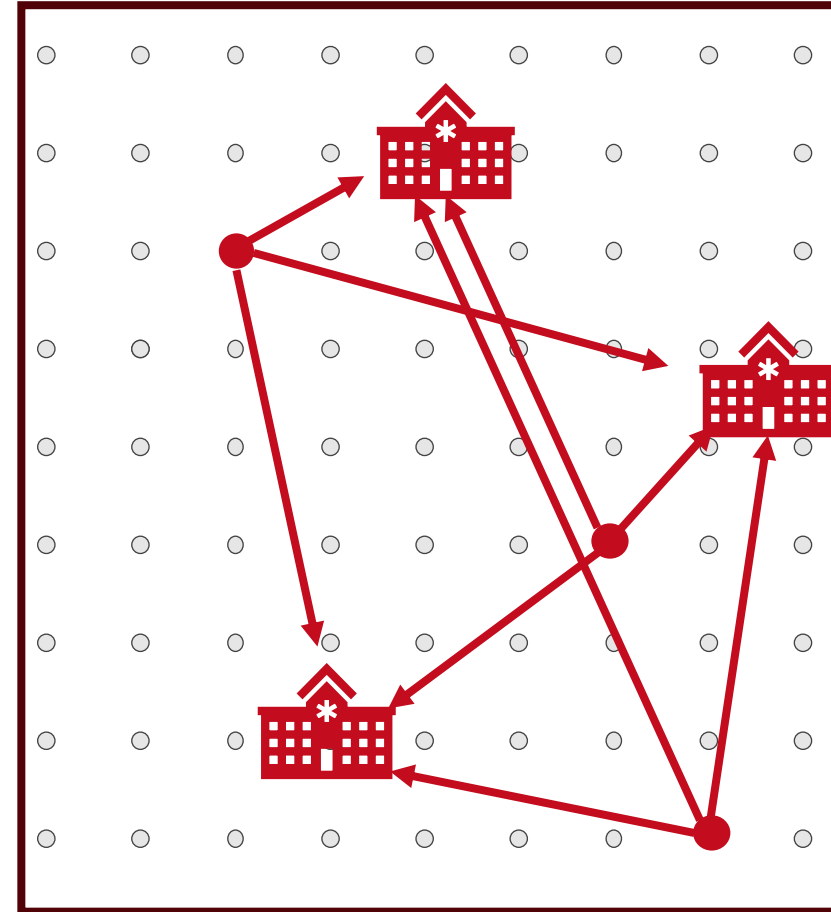


# Observational DRT Service Probe

“Realistic” Demand for DRT

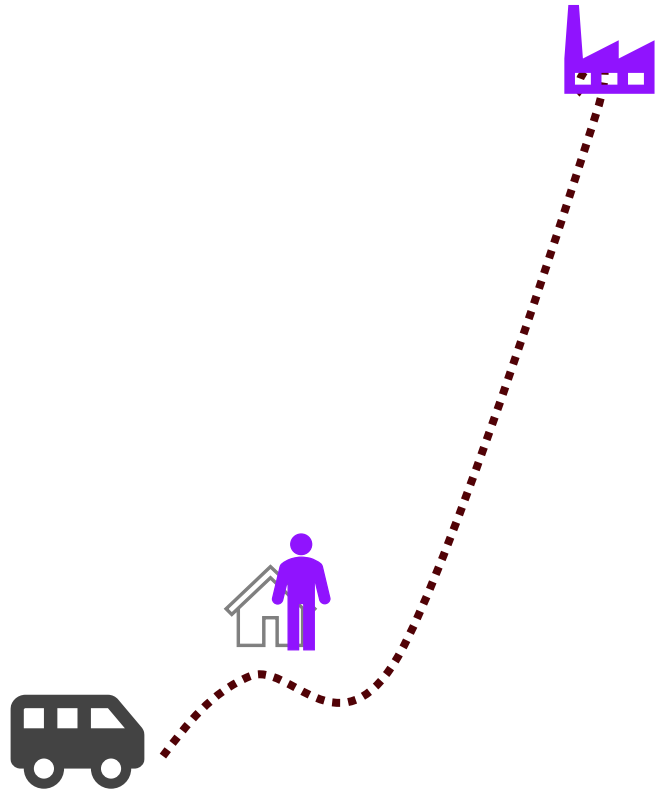


“Unrealistic” Demand needed for Accessibility Calculation

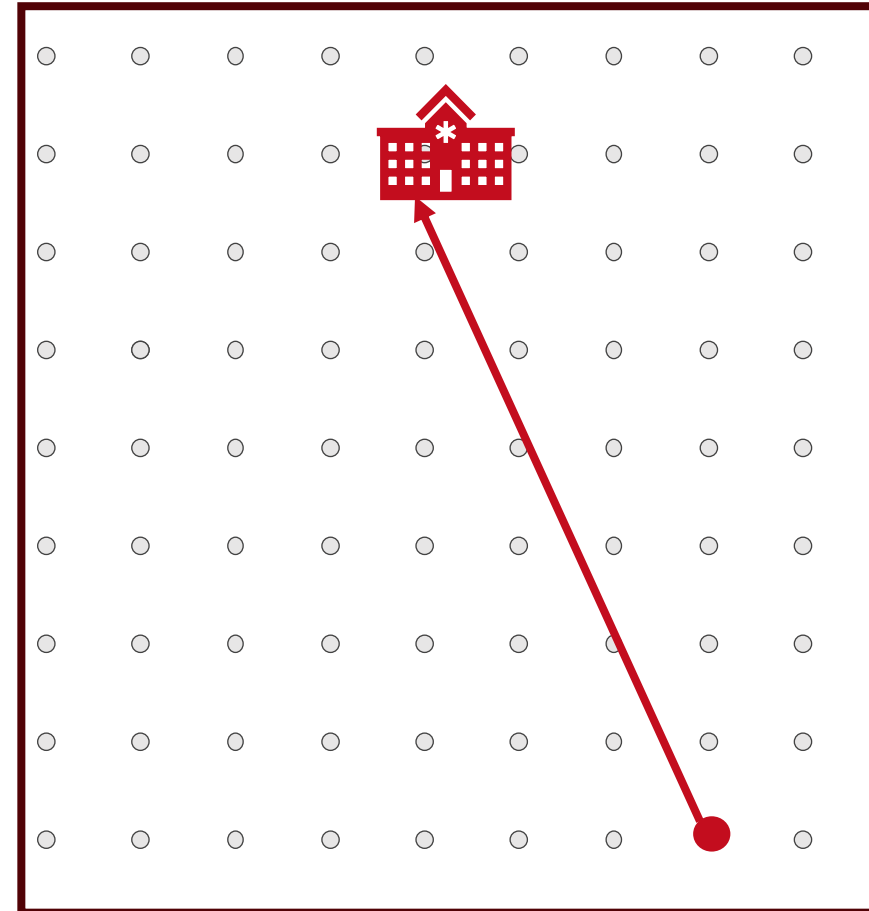


# Observational DRT Service Probe

“Realistic” Demand for DRT

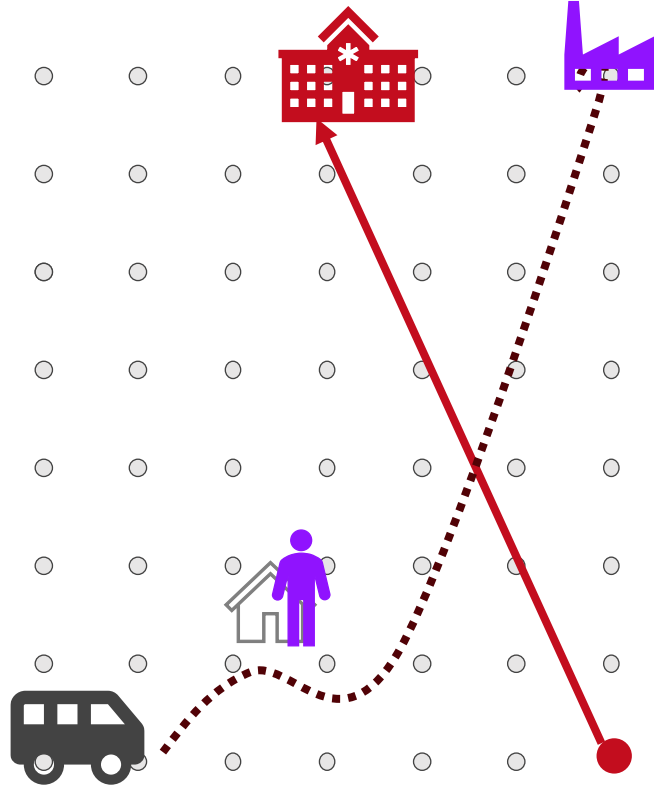


“Unrealistic” Demand needed for Accessibility Calculation



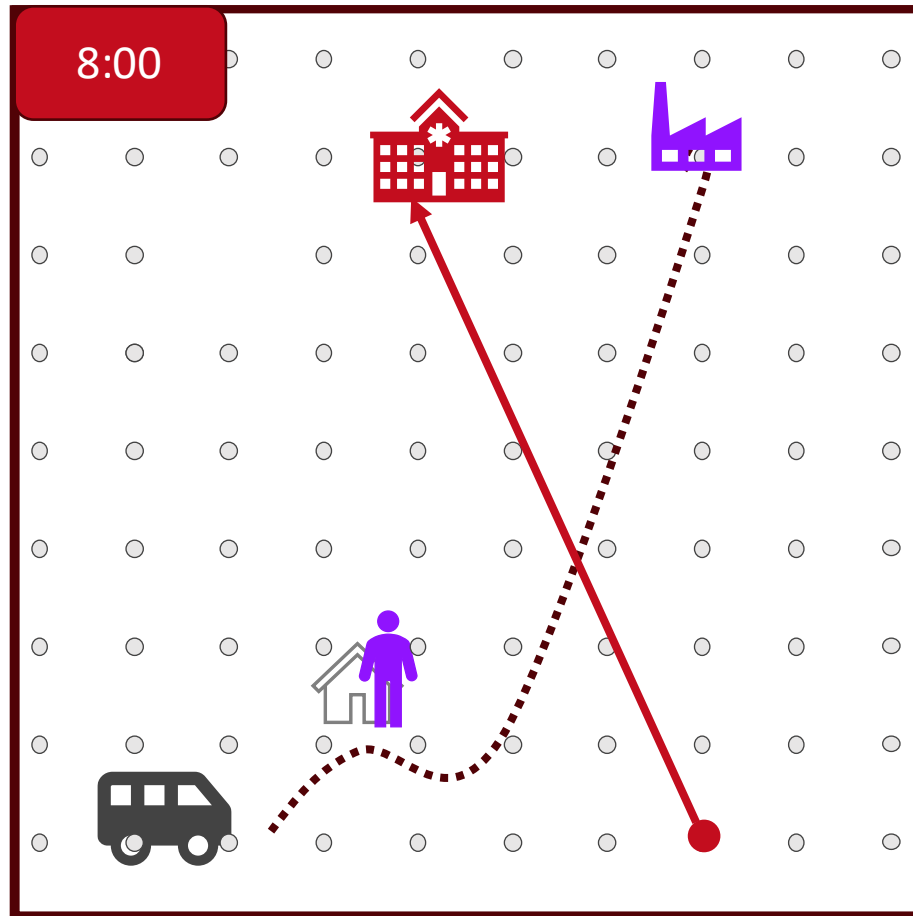
# Observational DRT Service Probe

“Realistic” Demand for DRT



“Unrealistic” Demand needed for Accessibility Calculation

# Observational DRT Service Probe



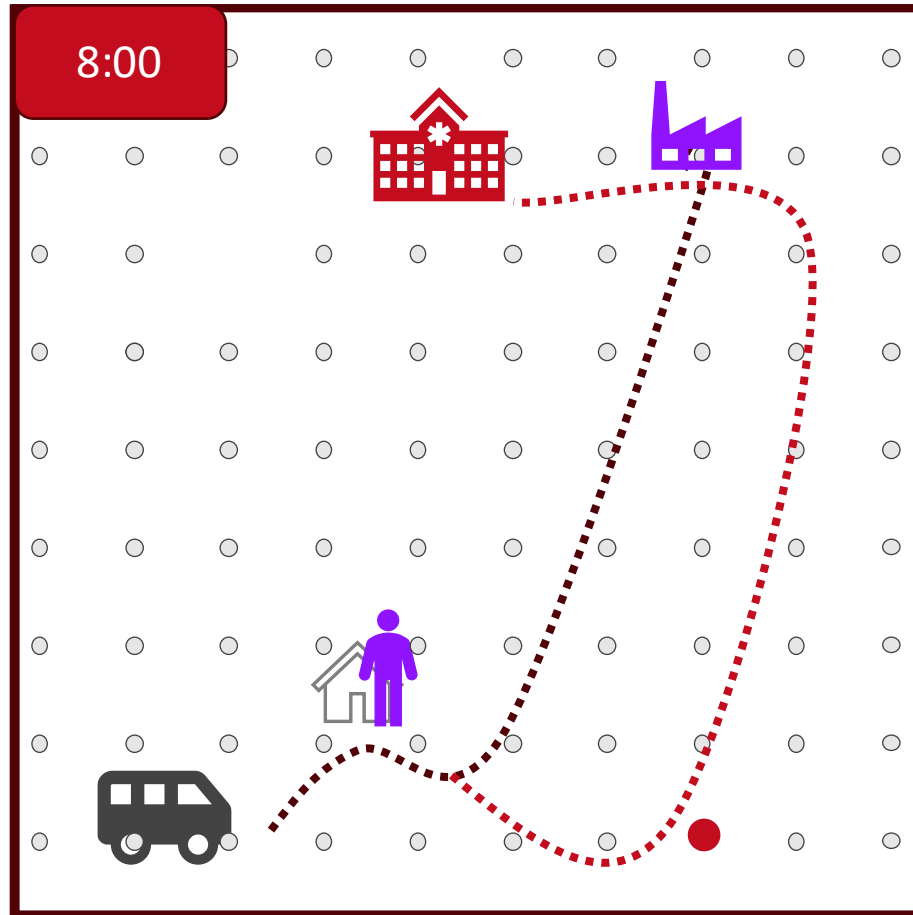
**Purpose:** To query DRT LOS for all combinations of ORIGIN x DESTINATION x TIME without affecting DRT's delicate balance between supply and demand

## Methodology:

During the last iteration of a MATSim simulation,

- FOR EACH probe time
  - Pause the simulation
    - FOR EACH origin–destination pair
      - Request a trip from the DRT dispatcher
      - Save the estimated wait time and detour factor
      - !!Do not schedule the trip!!
  - Resume Simulation

# Observational DRT Service Probe



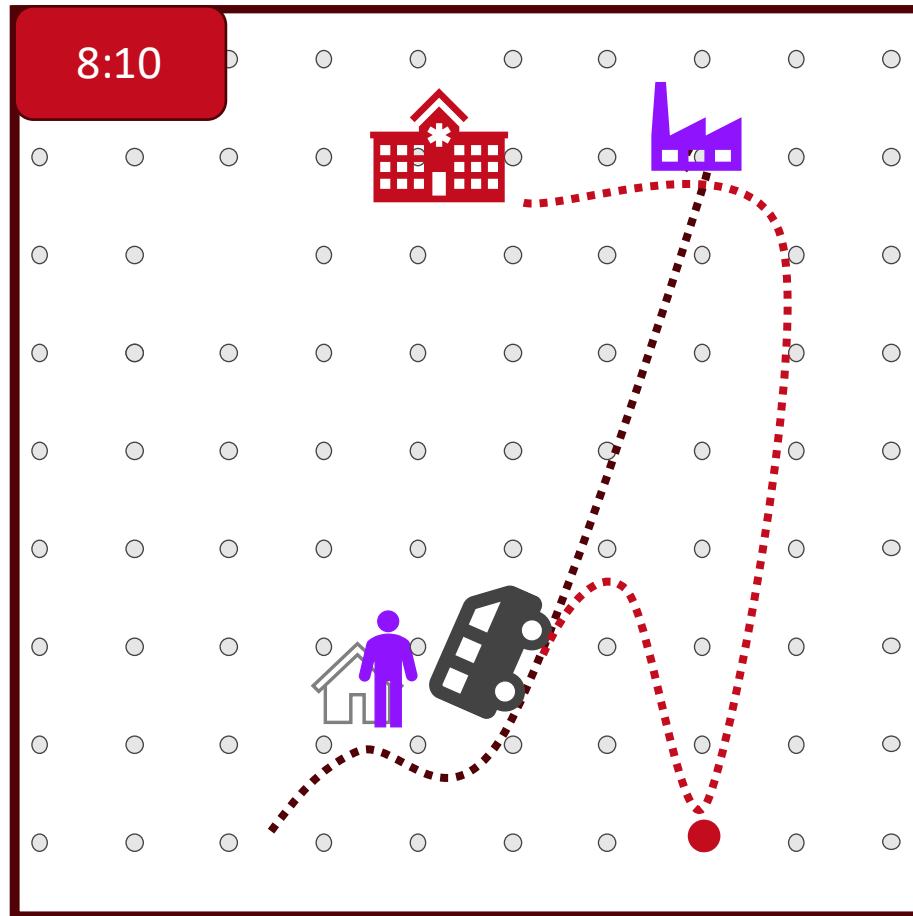
**Purpose:** To query DRT LOS for all combinations of ORIGIN x DESTINATION x TIME without affecting DRT's delicate balance between supply and demand

## Methodology:

During the last iteration of a MATSim simulation,

- FOR EACH probe time
  - Pause the simulation
    - FOR EACH origin–destination pair
      - Request a trip from the DRT dispatcher
      - Save the estimated wait time and detour factor
      - !!Do not schedule the trip!!
  - Resume Simulation

# Observational DRT Service Probe



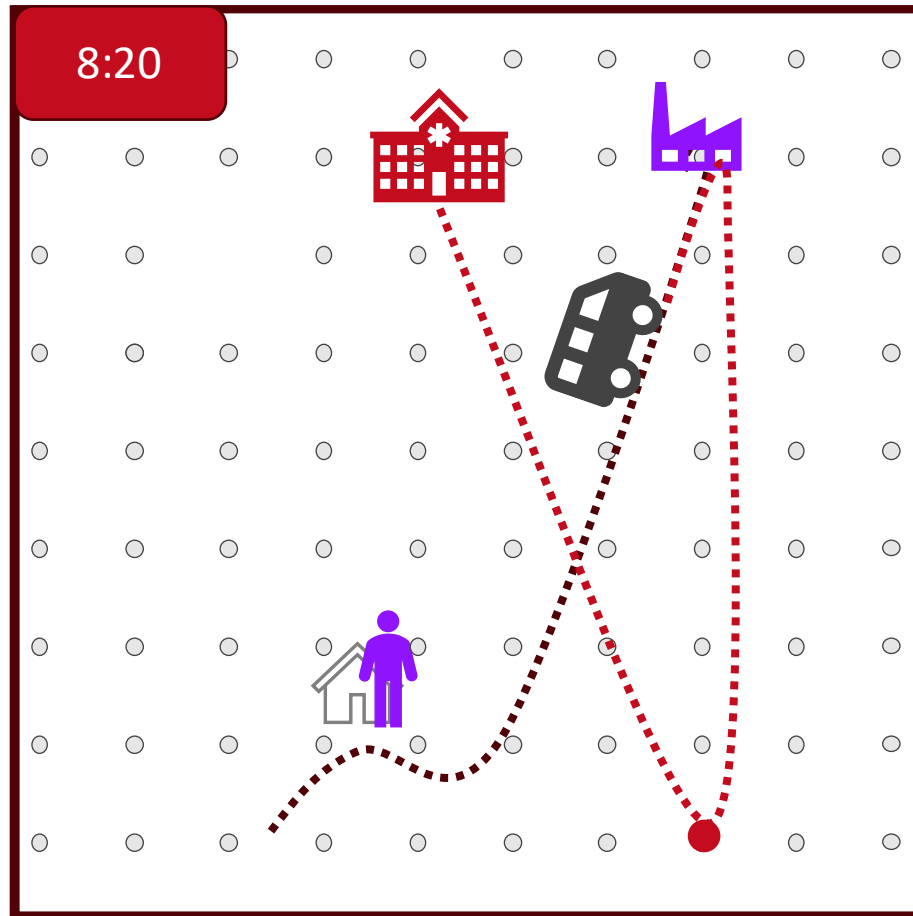
**Purpose:** To query DRT LOS for all combinations of ORIGIN x DESTINATION x TIME without affecting DRT's delicate balance between supply and demand

## Methodology:

During the last iteration of a MATSim simulation,

- FOR EACH probe time
  - Pause the simulation
    - FOR EACH origin–destination pair
      - Request a trip from the DRT dispatcher
      - Save the estimated wait time and detour factor
      - !!Do not schedule the trip!!
  - Resume Simulation

# Observational DRT Service Probe

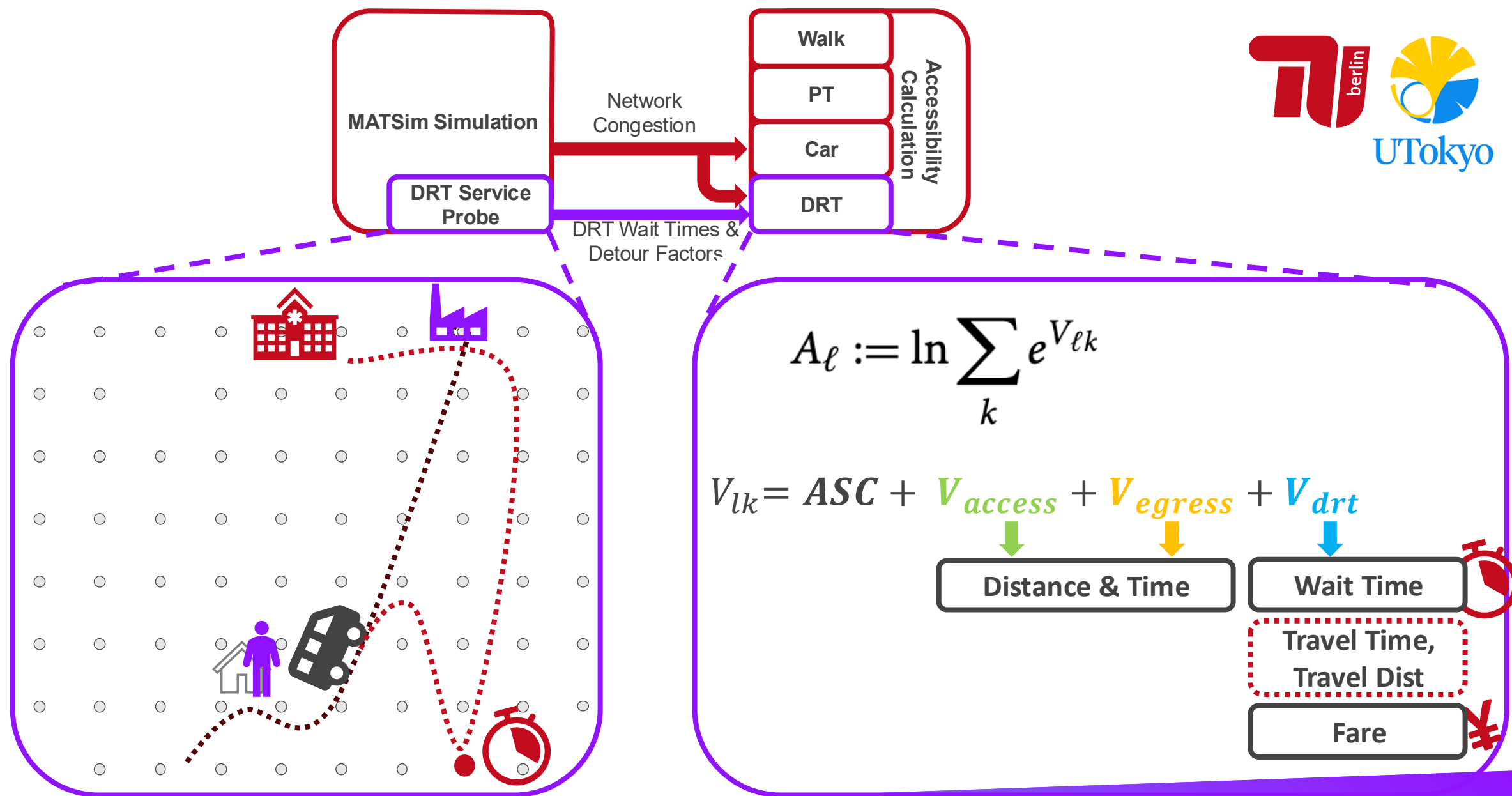


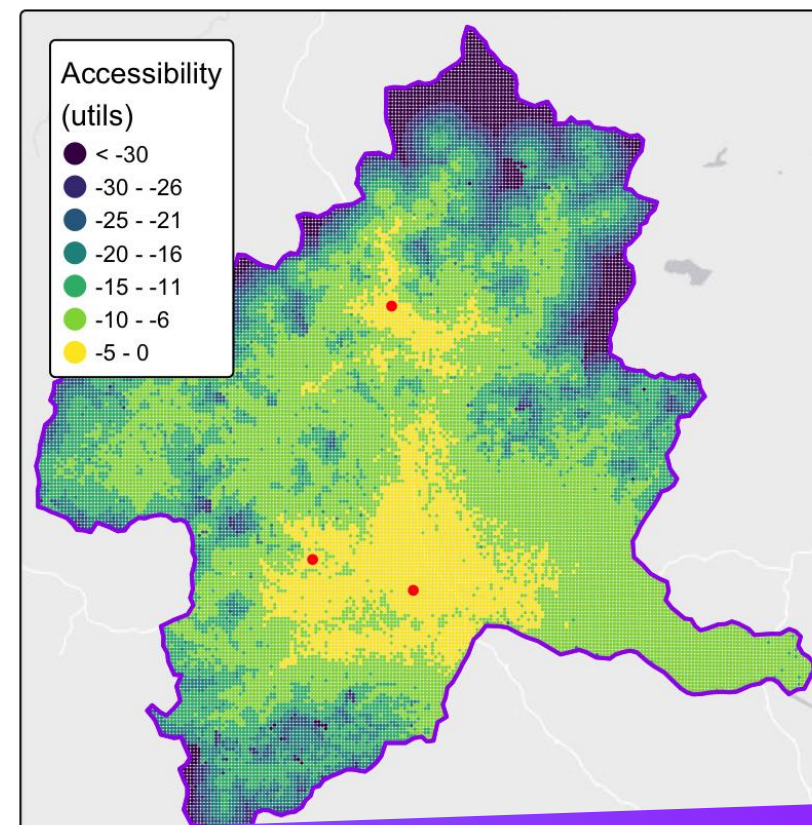
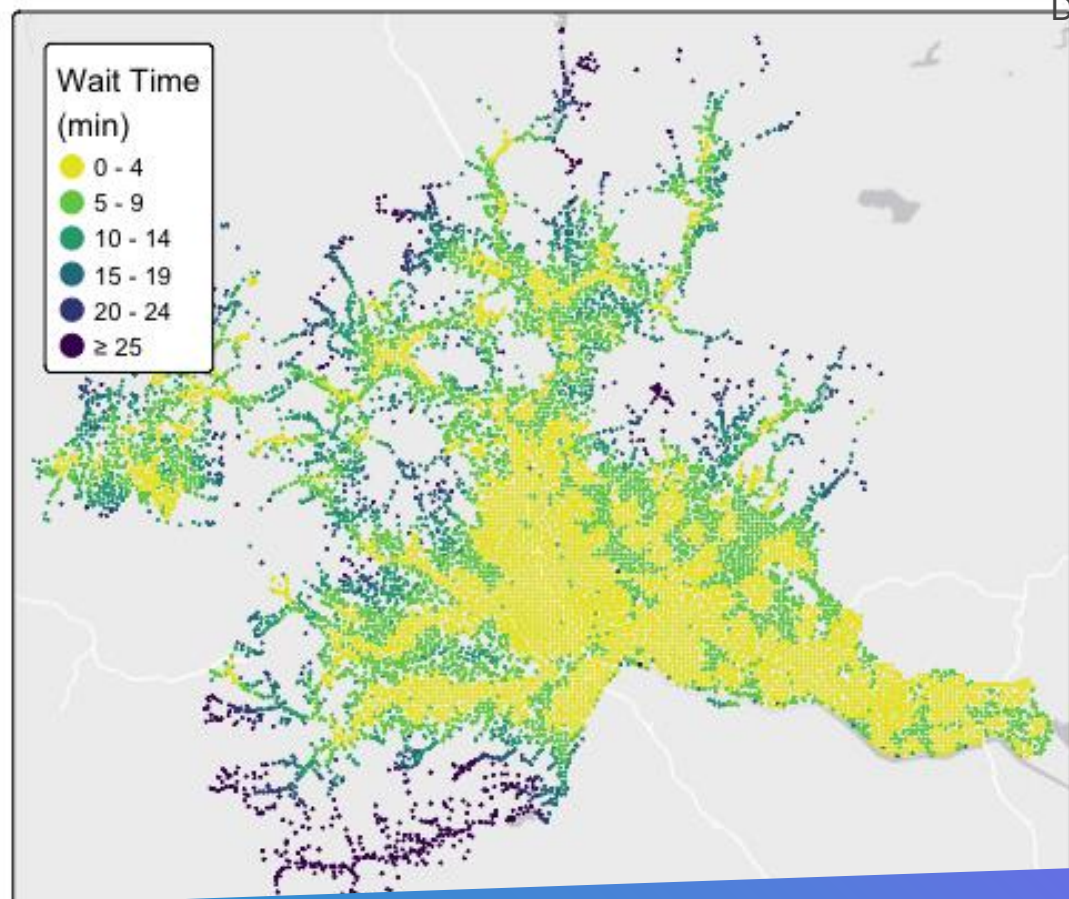
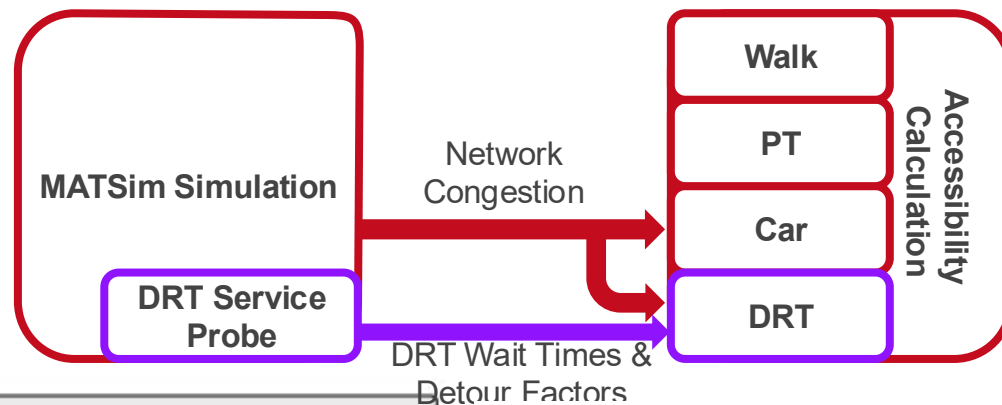
**Purpose:** To query DRT LOS for all combinations of ORIGIN x DESTINATION x TIME without affecting DRT's delicate balance between supply and demand

## Methodology:

During the last iteration of a MATSim simulation,

- FOR EACH probe time
  - Pause the simulation
    - FOR EACH origin–destination pair
      - Request a trip from the DRT dispatcher
      - Save the estimated wait time and detour factor
      - !!Do not schedule the trip!!
  - Resume Simulation





# Person-Based Accessibility

Introduction



DRT  
Accessibility

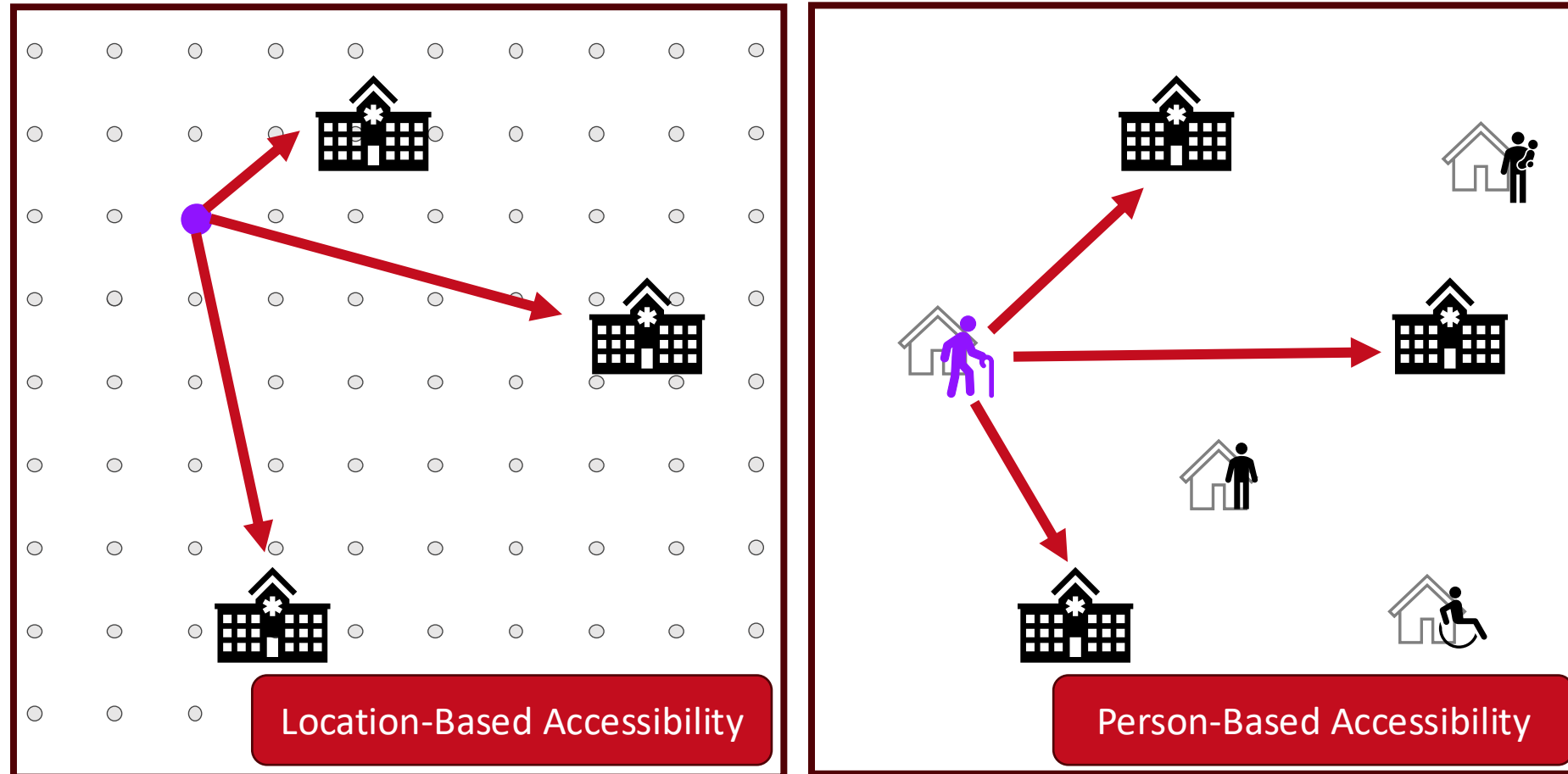


Person-based  
Accessibility

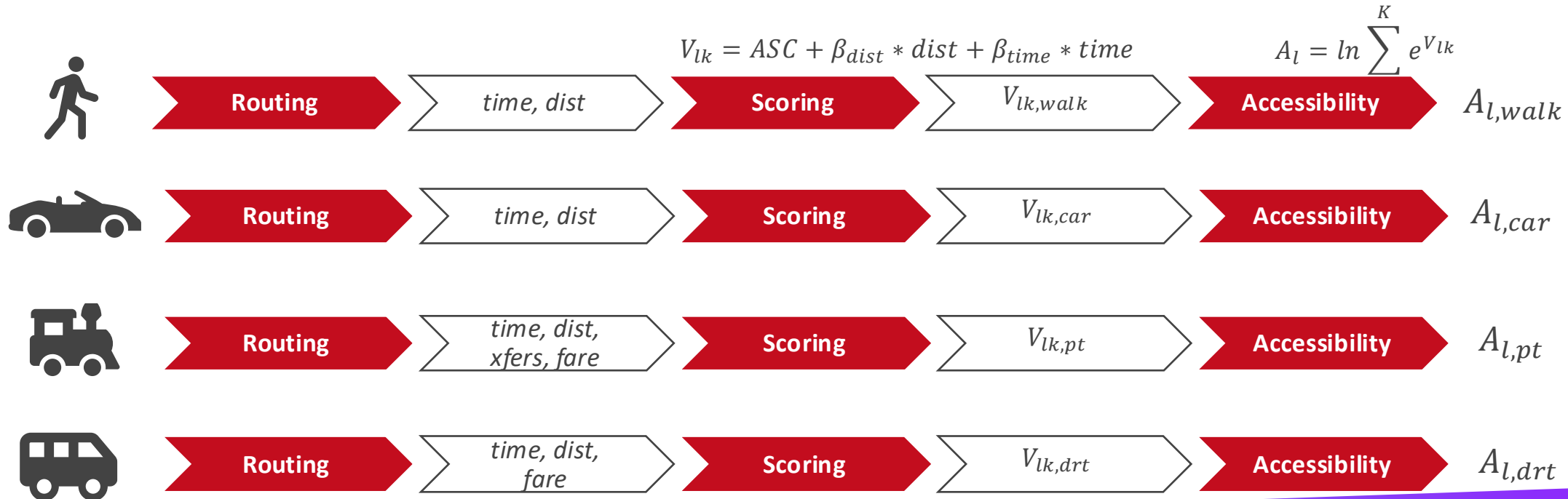
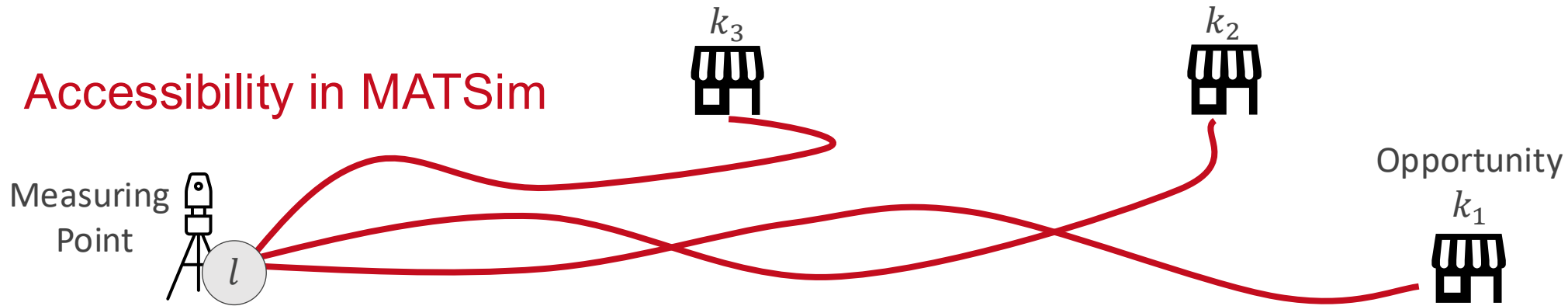


Discussion

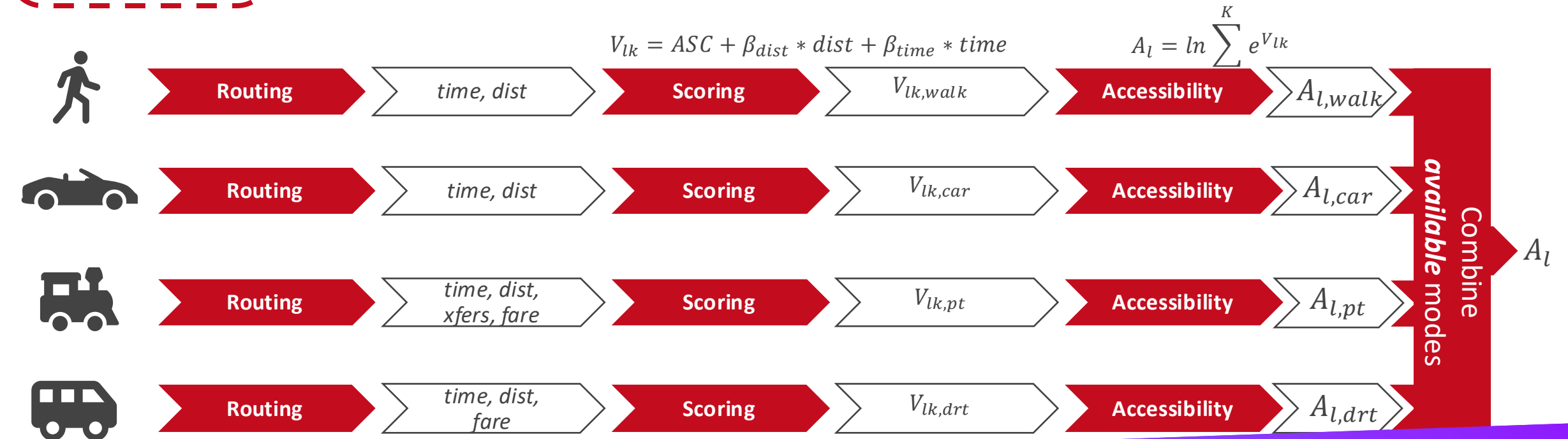
# Person-Based Accessibility



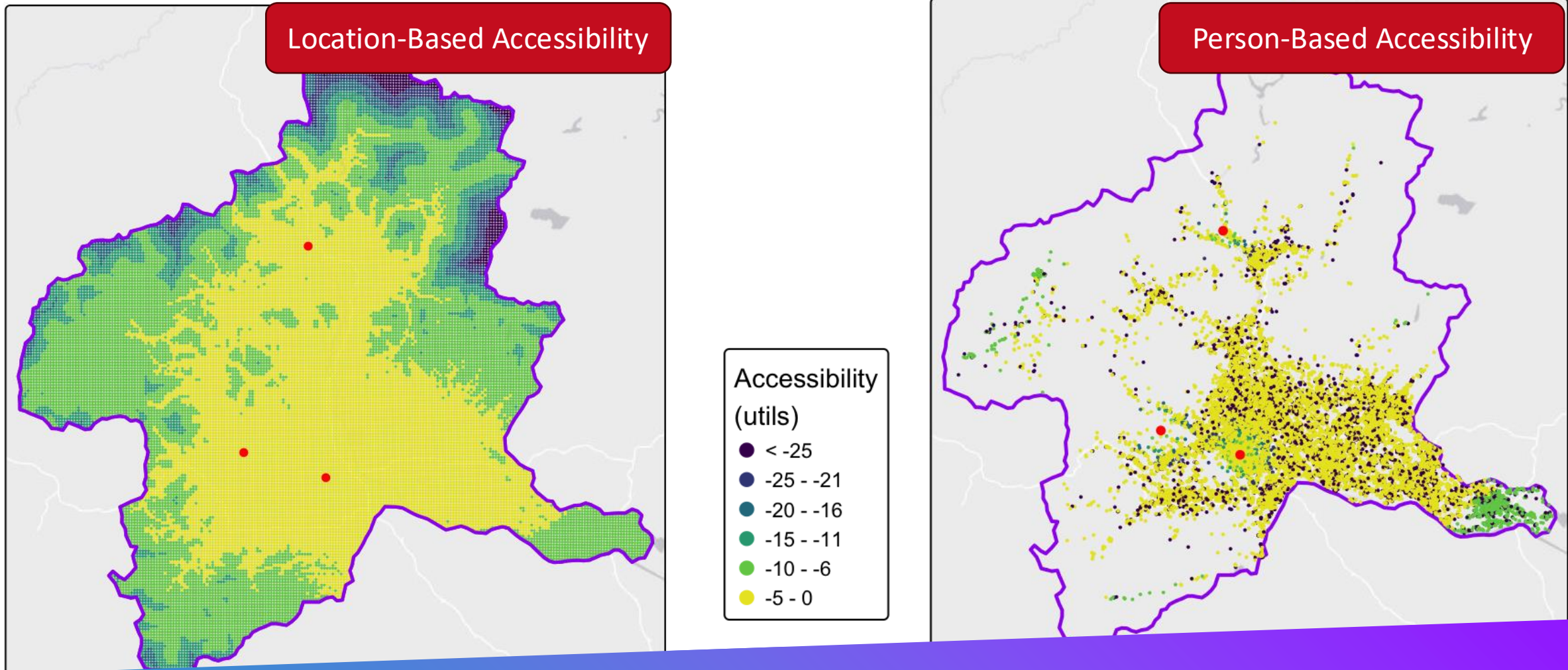
# Accessibility in MATSim



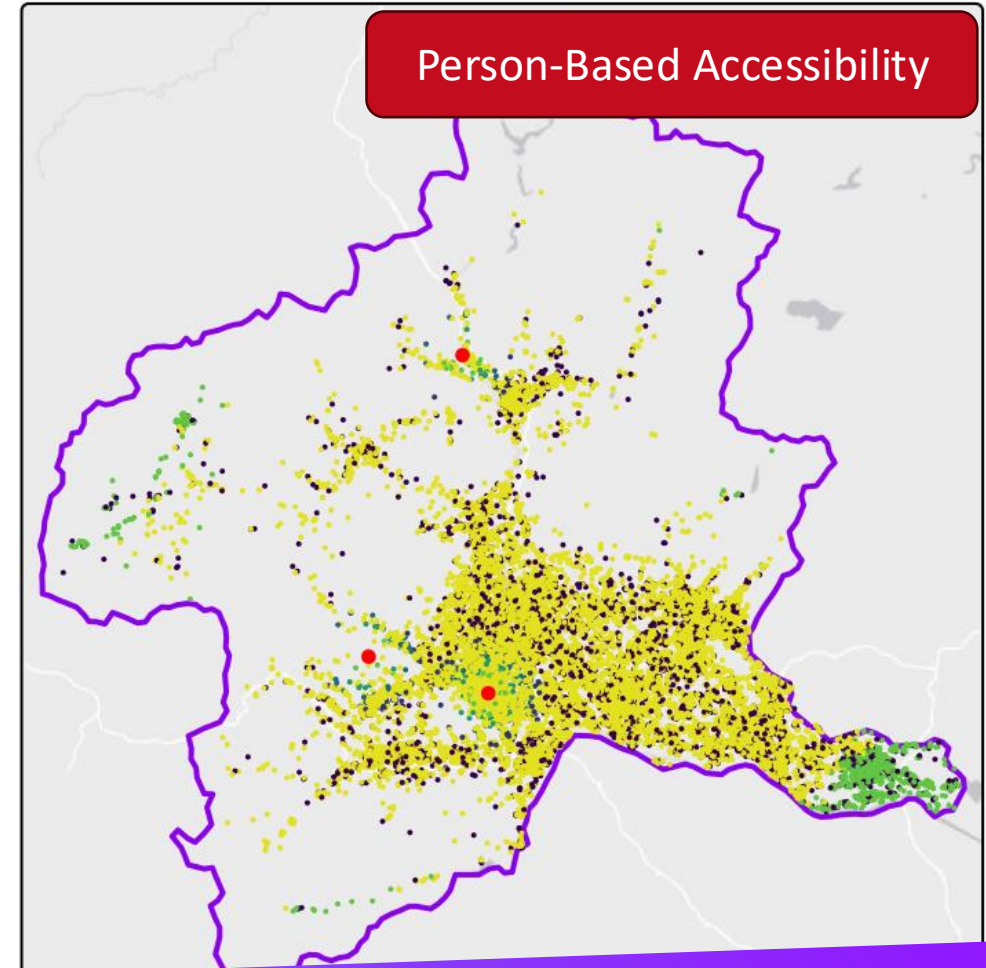
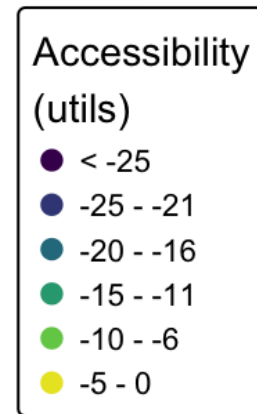
# Accessibility in MATSim



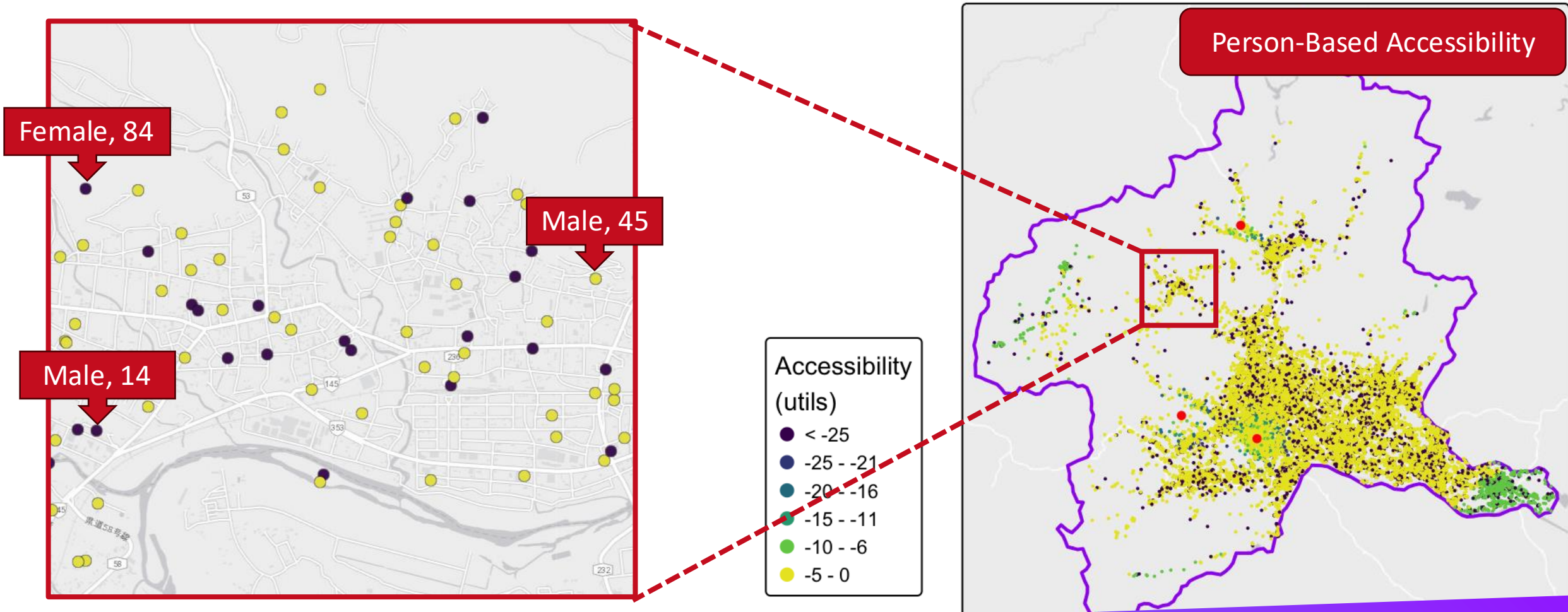
# Location-Based vs Person-Based Accessibility (Car + Walk)



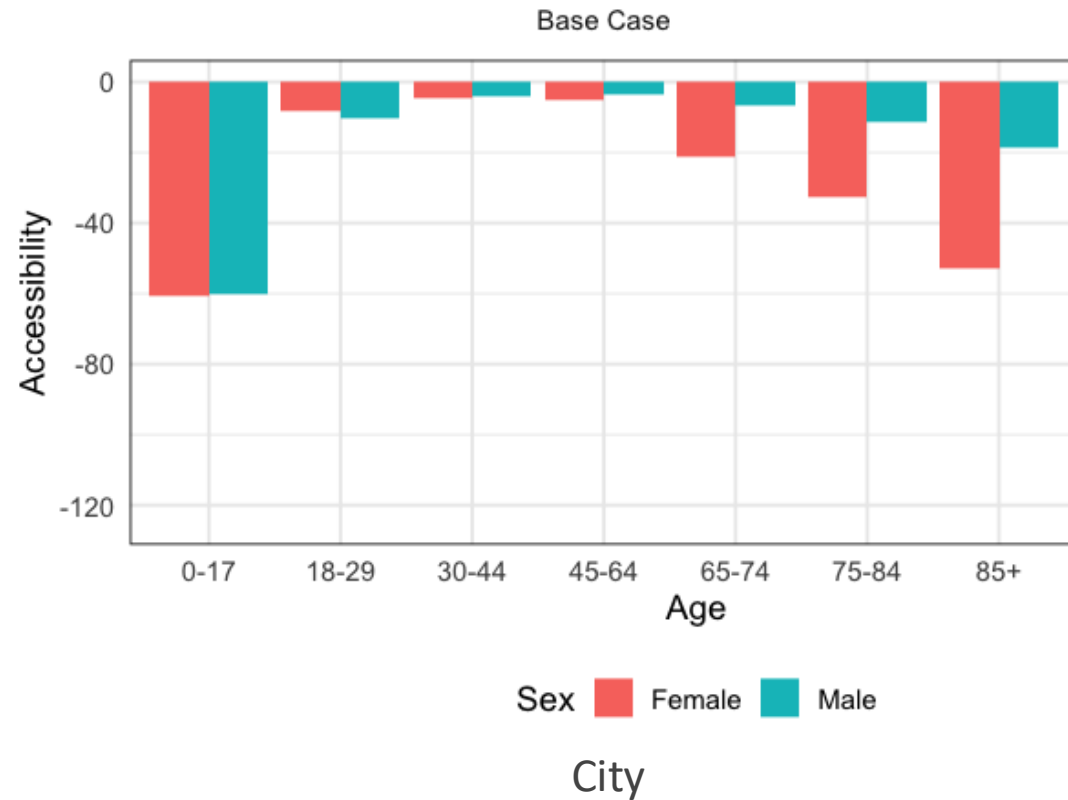
# Person-Based Accessibility to Shinkansen Stations



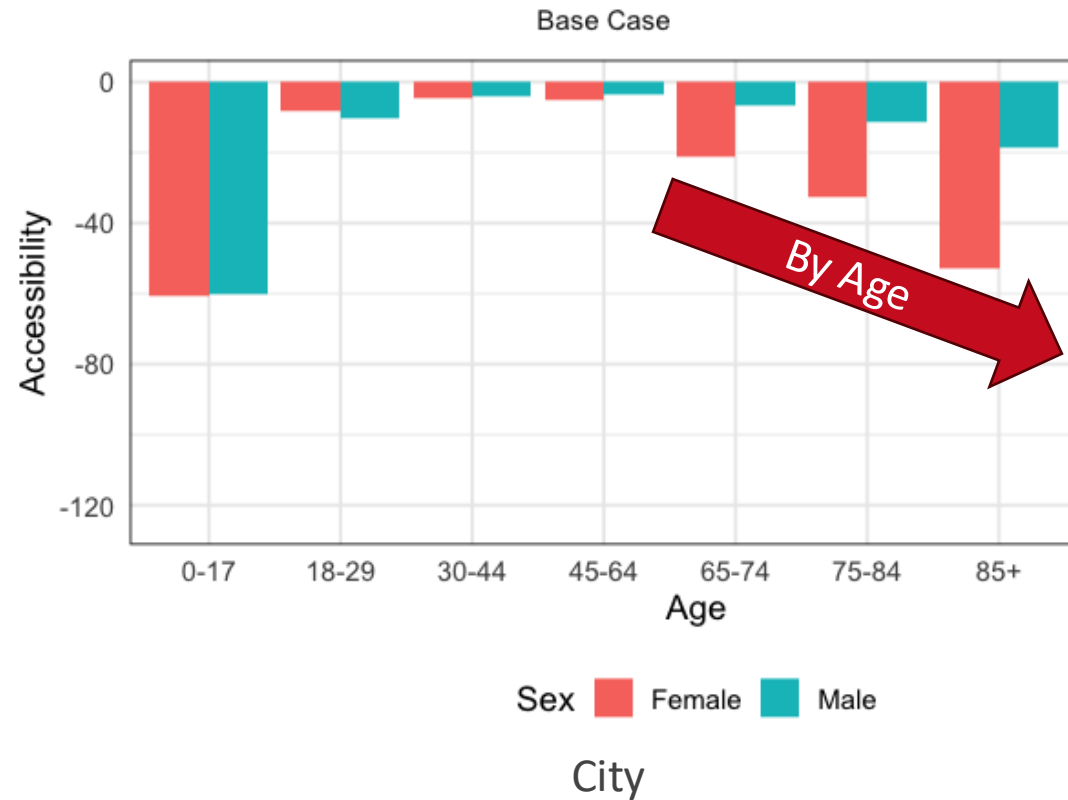
# Person-Based Accessibility to Shinkansen Stations



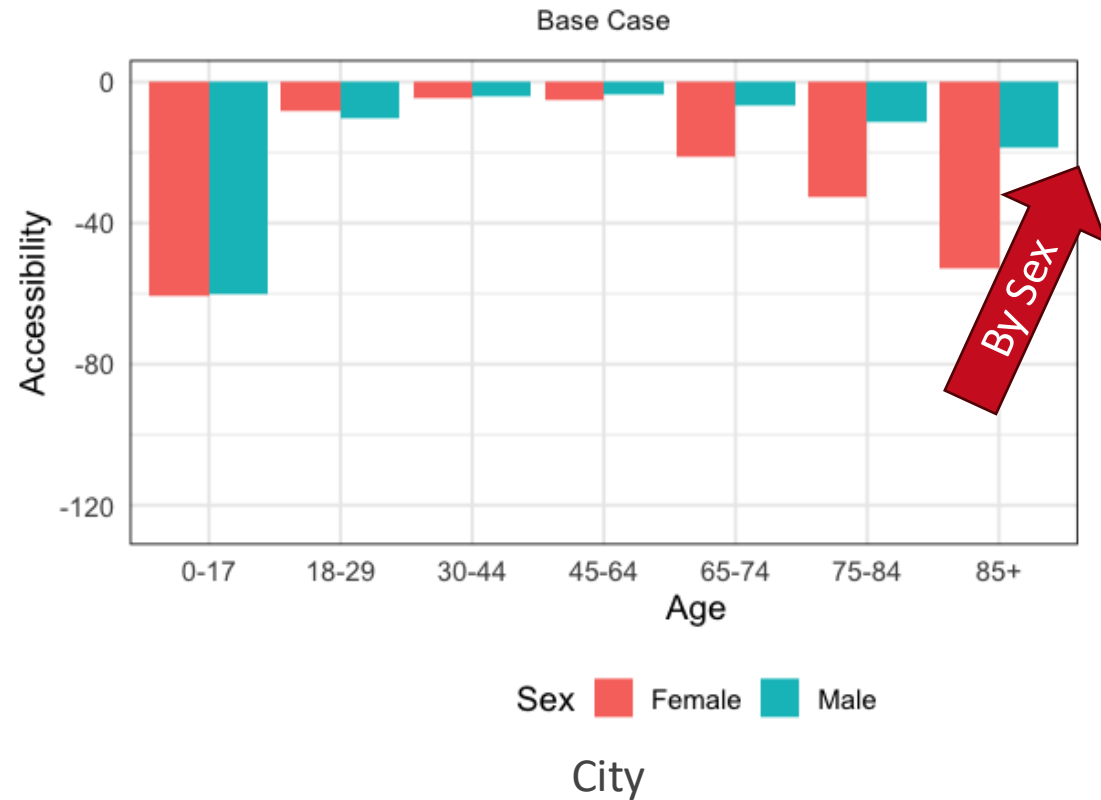
# Person-Based Accessibility to Shinkansen Stations



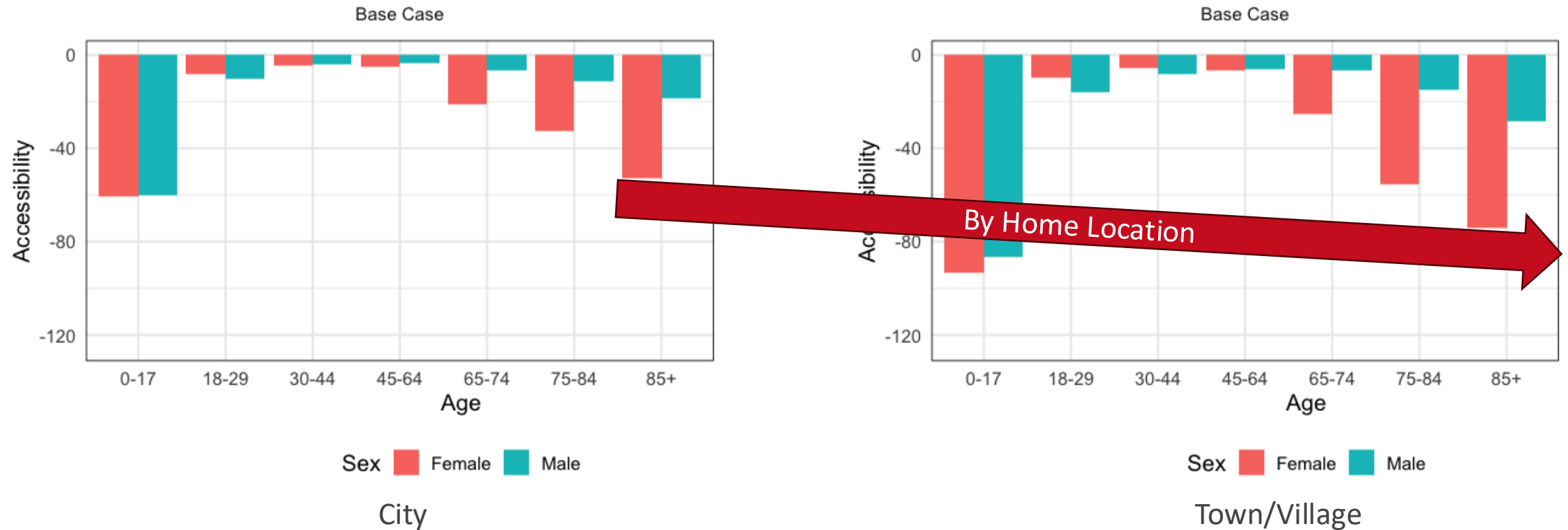
# Person-Based Accessibility to Shinkansen Stations



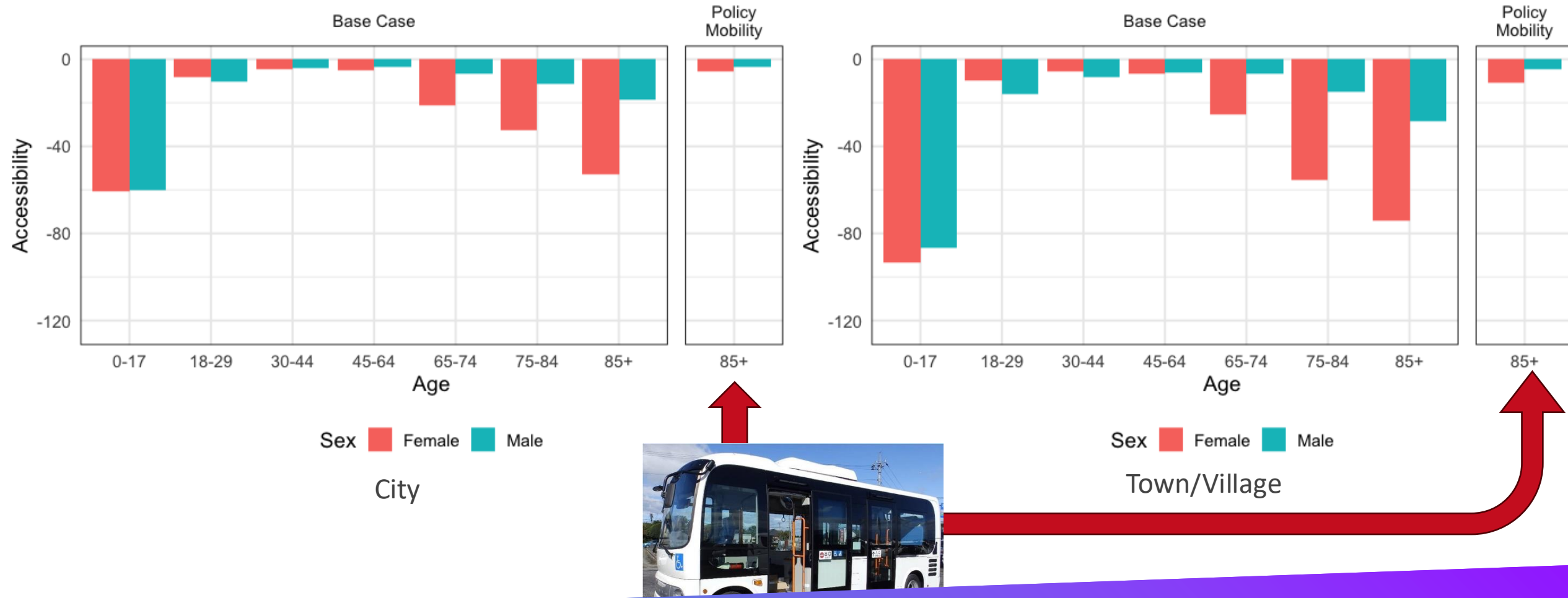
# Person-Based Accessibility to Shinkansen Stations



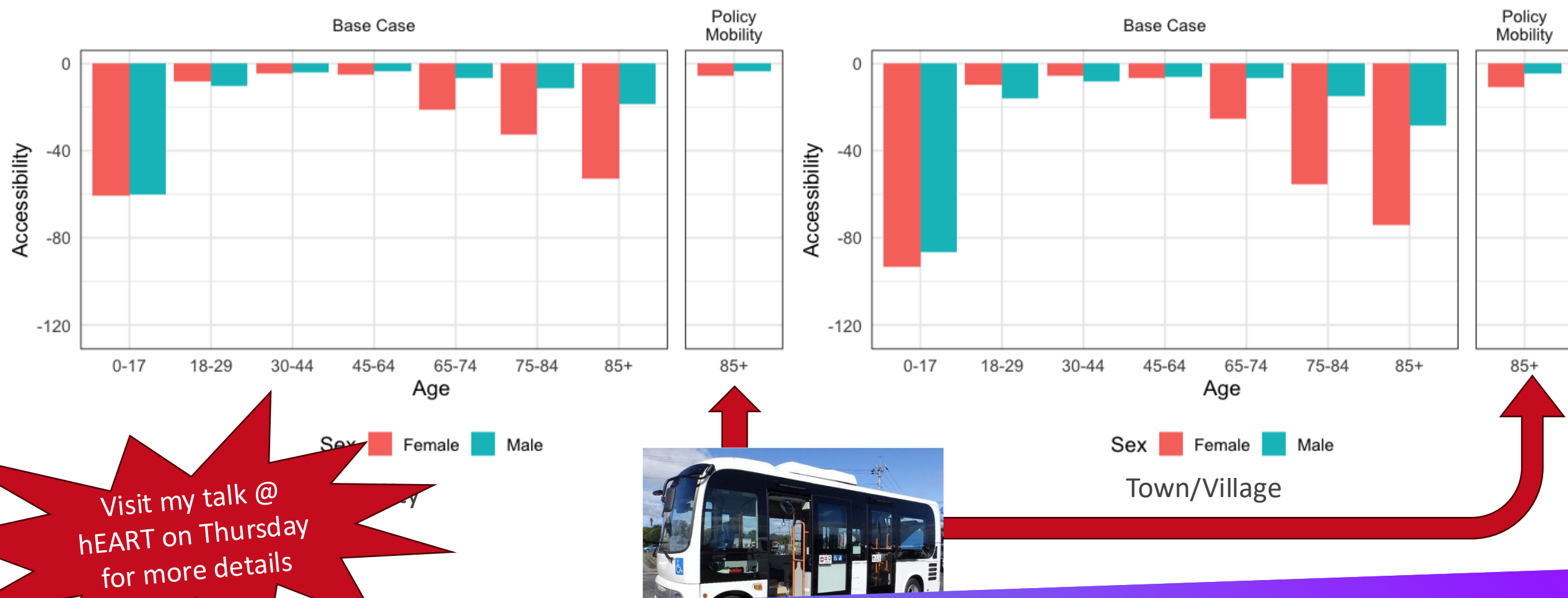
# Person-Based Accessibility to Shinkansen Stations



# Person-Based Accessibility to Shinkansen Stations with DRT



# Person-Based Accessibility to Shinkansen Stations with DRT



# Discussion

Introduction



DRT  
Accessibility



Person-based  
Accessibility



Discussion

# Outlook: Merge Pull Requests

## DRT Accessibility

 **feat(drt|accessibility): DRT Level of Service Output + Integration into DRT Accessibility**

#5060 · jakobrehmann opened on Jul 14 Member ·  [54/54](#)

 **feat(accessibility): DRT Accessibility**

#3626 · jakobrehmann opened on Dec 4, 2024 · Closed on Jan 12 Member ·  [53/53](#)

 13

## Person-Based Accessibility

 **feat(accessibility): Person-based accessibility**

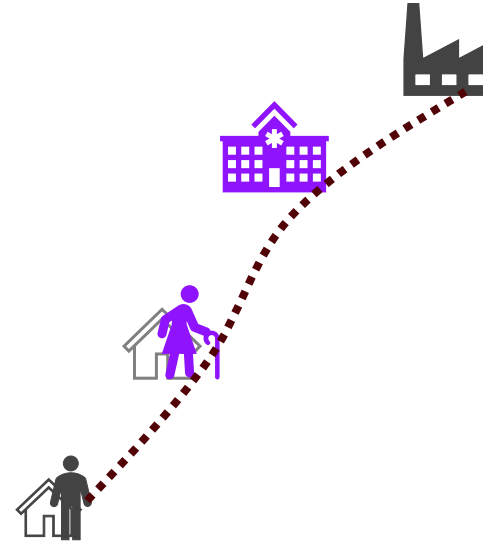
#4806 · jakobrehmann opened on Mar 16 Member ·  [52/52](#)

## Outlook: Accessibility of “ride” (= Car as Passenger)

- Ignoring ride trips in equity analysis of elderly people is clear limitation.
- Challenges for calculating ride accessibility:
  1. Which agents have access to ride trips?
  2. How to incorporate disutility of the driver?

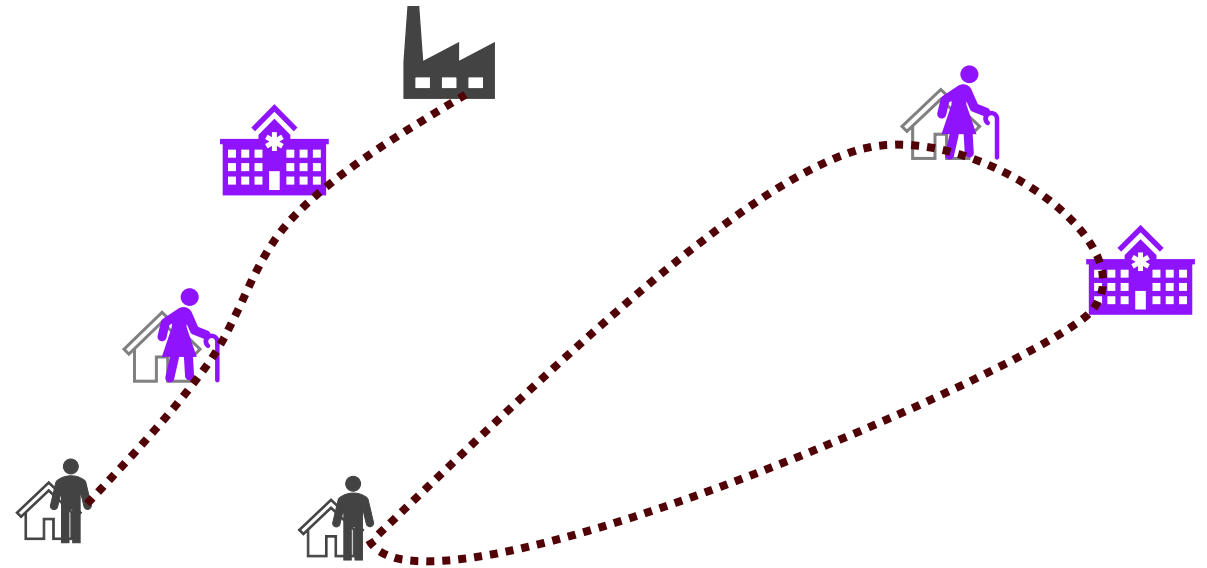
## Outlook: Accessibility of “ride” (= Car as Passenger)

- Ignoring ride trips in equity analysis of elderly people is clear limitation.
- Challenges for calculating ride accessibility:
  1. Which agents have access to ride trips?
  2. How to incorporate disutility of the driver?

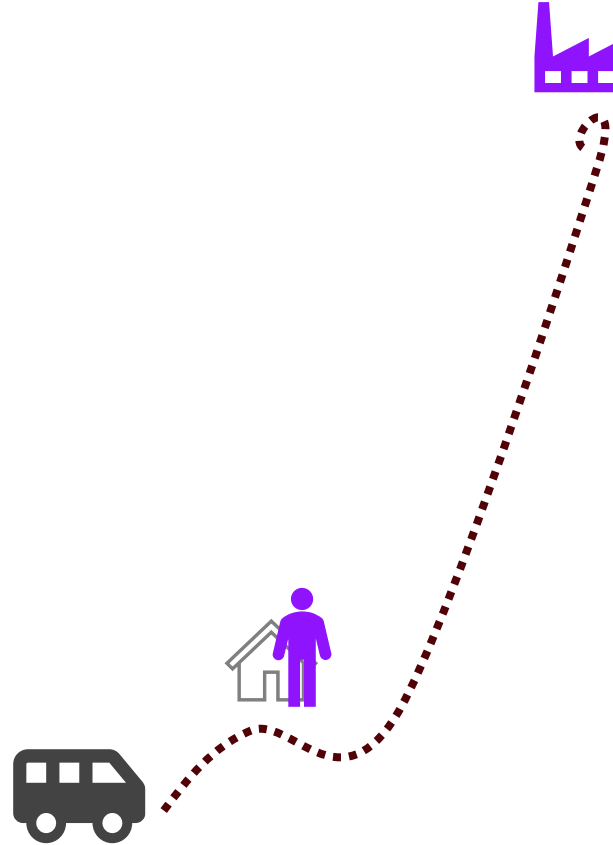


## Outlook: Accessibility of “ride” (= Car as Passenger)

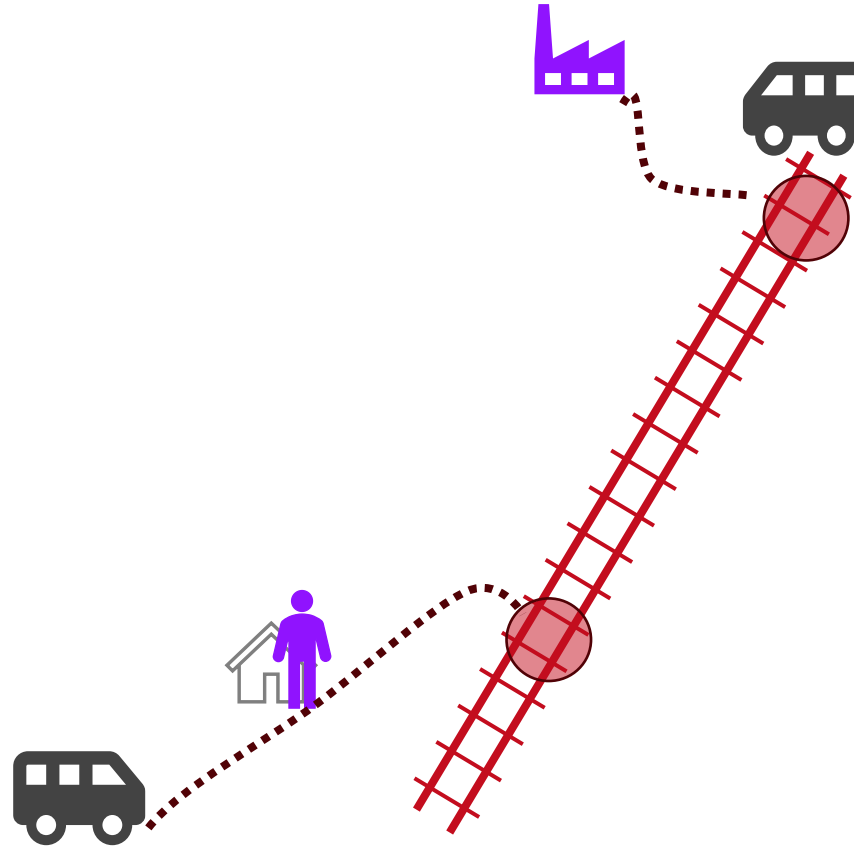
- Ignoring ride trips in equity analysis of elderly people is clear limitation.
- Challenges for calculating ride accessibility:
  1. Which agents have access to ride trips?
  2. How to incorporate disutility of the driver?



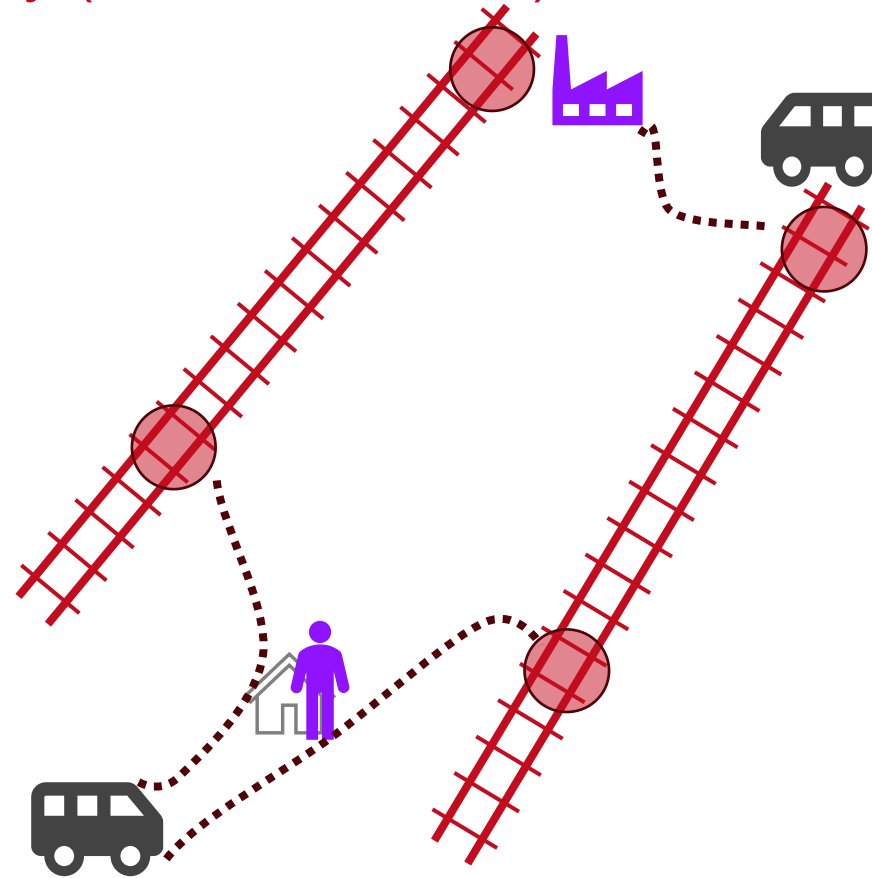
## Outlook: Intermodal Accessibility (DRT as Feeder)



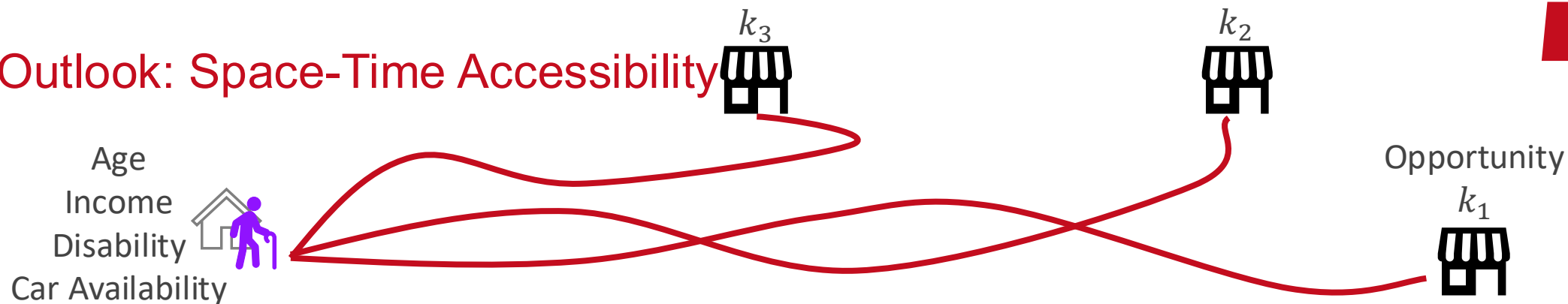
## Outlook: Intermodal Accessibility (DRT as Feeder)



## Outlook: Intermodal Accessibility (DRT as Feeder)



# Outlook: Space-Time Accessibility

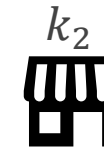


Space-Time Accessibility:

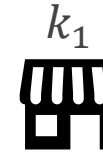
a) From Home

# Outlook: Space-Time Accessibility

Age  
Income  
Disability  
Car Availability



Opportunity



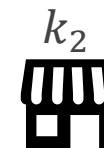
Space-Time Accessibility:

a) From Home



# Outlook: Space-Time Accessibility

Age  
Income  
Disability  
Car Availability

Opportunity



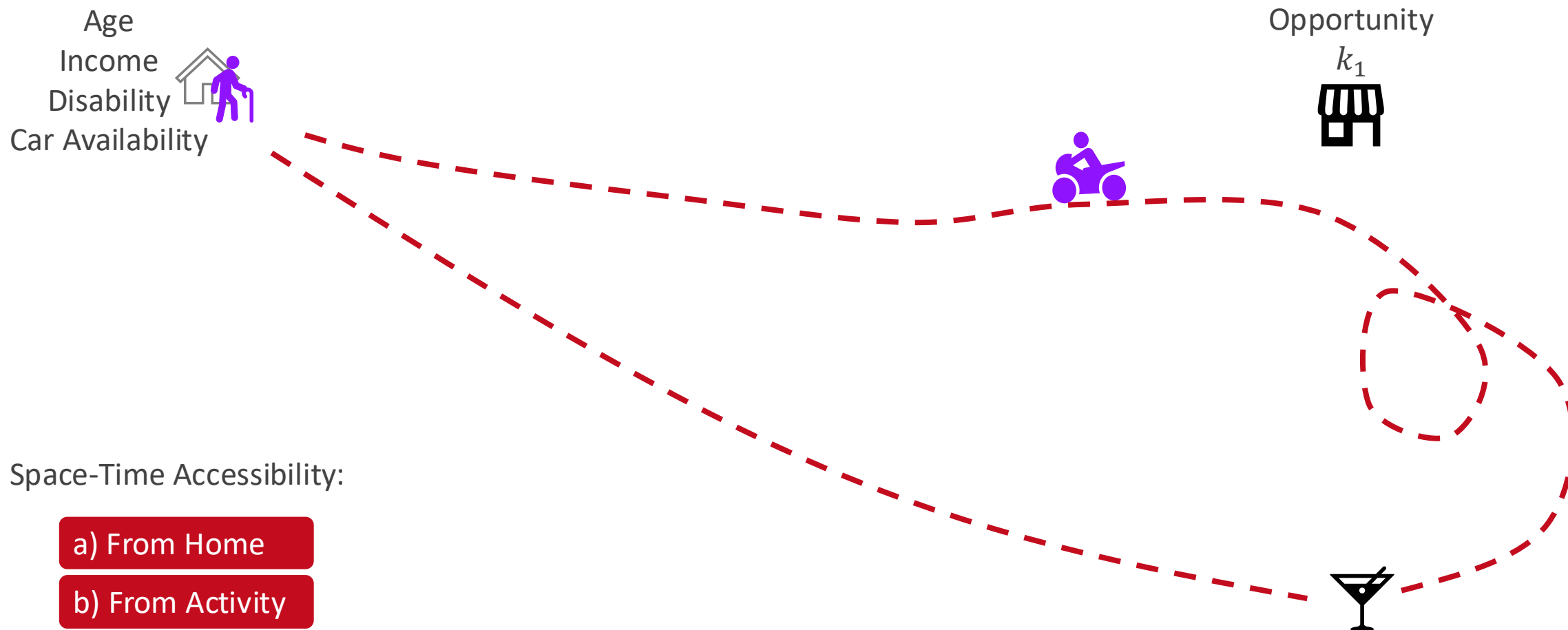
Space-Time Accessibility:

a) From Home

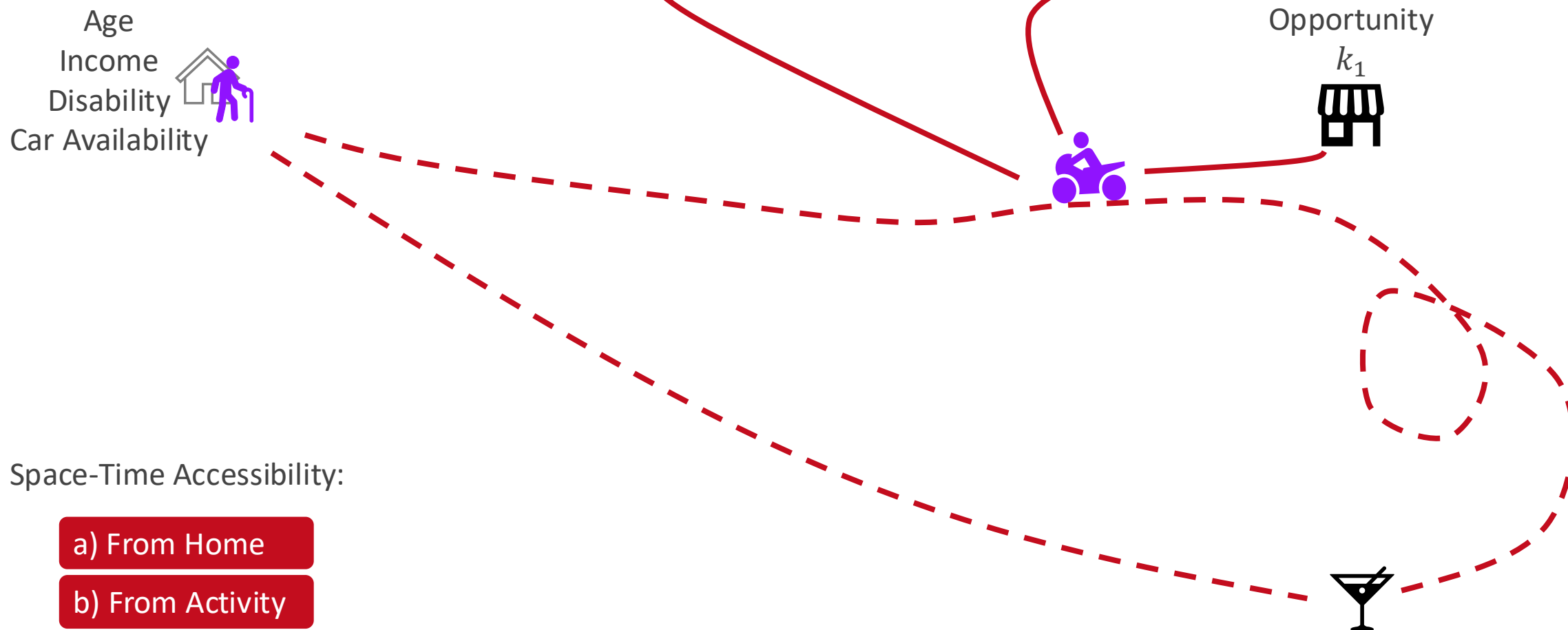
b) From Activity



# Outlook: Space-Time Accessibility



# Outlook: Space-Time Accessibility



a) From Home

b) From Activity

c) From Route

# Thank you!

## Questions?

**Feel free to write me!**

Email: [rehmann@vsp.tu-berlin.de](mailto:rehmann@vsp.tu-berlin.de)

LinkedIn: [jakob-rehmann](#)

((Or just chat with me in the break ☺ ))